

Course Catalogue

Complete Course & Lesson Plan

|               |          |         |               |
|---------------|----------|---------|---------------|
| 39            | 7        | 32      | 1365          |
| Total Courses | Complete | Pending | Total Lessons |

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# Programming Languages

## Java Mastery

■ Pending

Slug: **java** · Difficulty: **Intermediate** · Duration: **40h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Java, OOP, JVM, Concurrency, Streams, Java 21

### Module 1

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 01 | Introduction to Java and the JVM Ecosystem       | ■ Reading  | 30 min · 75 XP  |
| 02 | Variables, Data Types, and Operators             | ■ Reading  | 30 min · 75 XP  |
| 03 | Control Flow: If-Else, Switch, and Ternary       | ■ Reading  | 30 min · 75 XP  |
| 04 | Loops: For, While, Do-While, and Enhanced For    | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Calculator and Number Games | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 06 | Classes, Objects, and Constructors             | ■ Reading  | 35 min · 75 XP  |
| 07 | Encapsulation and Access Modifiers             | ■ Reading  | 35 min · 75 XP  |
| 08 | Inheritance and Method Overriding              | ■ Reading  | 35 min · 75 XP  |
| 09 | Polymorphism, Interfaces, and Abstract Classes | ■ Reading  | 35 min · 75 XP  |
| 10 | OOP Practice: Building a Banking System        | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 11 | Arrays and the Java Collections Framework   | ■ Reading  | 35 min · 75 XP  |
| 12 | Generics: Type Safety and Wildcards         | ■ Reading  | 35 min · 75 XP  |
| 13 | Exception Handling and Custom Exceptions    | ■ Reading  | 35 min · 75 XP  |
| 14 | String Manipulation and Regular Expressions | ■ Reading  | 35 min · 75 XP  |
| 15 | Core APIs Assessment and Practice           | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                |           |                 |
|----|------------------------------------------------|-----------|-----------------|
| 16 | Lambda Expressions and Functional Interfaces   | ■ Reading | 35 min · 75 XP  |
| 17 | Streams API: Filter, Map, Reduce               | ■ Reading | 35 min · 75 XP  |
| 18 | Optional: Null-Safe Programming                | ■ Reading | 35 min · 75 XP  |
| 19 | Method References and Composing Functions      | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: E-Commerce Product Catalog | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 21 | Threads, Runnable, and the Executor Framework       | ■ Reading  | 35 min · 75 XP  |
| 22 | Synchronization, Locks, and Concurrent Collections  | ■ Reading  | 35 min · 75 XP  |
| 23 | CompletableFuture and Async Programming             | ■ Reading  | 35 min · 75 XP  |
| 24 | Java 21 Virtual Threads and Structured Concurrency  | ■ Reading  | 35 min · 75 XP  |
| 25 | Concurrency Practice: Multi-threaded Data Processor | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 26 | I/O Streams, NIO.2, and File Handling              | ■ Reading  | 35 min · 75 XP  |
| 27 | Java Module System (JPMS)                          | ■ Reading  | 35 min · 75 XP  |
| 28 | Records, Sealed Classes, and Pattern Matching      | ■ Reading  | 35 min · 75 XP  |
| 29 | JVM Internals: Memory Model and Garbage Collection | ■ Reading  | 35 min · 75 XP  |
| 30 | Modern Java Features Practice                      | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                  |           |                |
|----|----------------------------------|-----------|----------------|
| 31 | Testing with JUnit 5 and Mockito | ■ Reading | 35 min · 75 XP |
|----|----------------------------------|-----------|----------------|

|    |                                                     |           |                 |
|----|-----------------------------------------------------|-----------|-----------------|
| 32 | Build Tools: Maven and Gradle                       | ■ Reading | 30 min · 75 XP  |
| 33 | Logging, Profiling, and Debugging Java Applications | ■ Reading | 35 min · 75 XP  |
| 34 | Performance Tuning and JVM Flags                    | ■ Reading | 35 min · 75 XP  |
| 35 | Capstone: Production-Ready Java Microservice        | ■ Project | 60 min · 150 XP |

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# Python for Everyone

■ Pending

Slug: **python** · Difficulty: **Beginner** · Duration: **28h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Python, Scripting, Automation, Data

## Module 1

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 01 | Introduction to Python: Why Python and How It Works | ■ Reading  | 30 min · 75 XP  |
| 02 | Variables, Data Types, and Basic Operations         | ■ Reading  | 30 min · 75 XP  |
| 03 | Strings: Formatting, Methods, and f-Strings         | ■ Reading  | 30 min · 75 XP  |
| 04 | Control Flow: If-Else, For, and While Loops         | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Python Scripting Challenges    | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 06 | Functions: Definition, Arguments, and Return Values | ■ Reading  | 35 min · 75 XP  |
| 07 | Lists, Tuples, Sets, and Dictionaries               | ■ Reading  | 35 min · 75 XP  |
| 08 | List Comprehensions and Generator Expressions       | ■ Reading  | 35 min · 75 XP  |
| 09 | File I/O: Reading, Writing, and CSV Handling        | ■ Reading  | 35 min · 75 XP  |
| 10 | Data Structures Practice: Grocery Store Manager     | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 11 | Object-Oriented Programming: Classes and Objects | ■ Reading  | 35 min · 75 XP  |
| 12 | Inheritance, Encapsulation, and Polymorphism     | ■ Reading  | 35 min · 75 XP  |
| 13 | Modules, Packages, and pip                       | ■ Reading  | 35 min · 75 XP  |
| 14 | Exception Handling and Debugging Techniques      | ■ Reading  | 35 min · 75 XP  |
| 15 | OOP Assessment: Library Management System        | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                    |           |                 |
|----|----------------------------------------------------|-----------|-----------------|
| 16 | Working with Dates, Times, and the datetime Module | ■ Reading | 30 min · 75 XP  |
| 17 | Regular Expressions with the re Module             | ■ Reading | 35 min · 75 XP  |
| 18 | Decorators and Context Managers                    | ■ Reading | 35 min · 75 XP  |
| 19 | Iterators, Generators, and Lazy Evaluation         | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Command-Line Task Manager      | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 21 | Virtual Environments and Dependency Management    | ■ Reading  | 30 min · 75 XP  |
| 22 | Working with JSON, XML, and APIs                  | ■ Reading  | 35 min · 75 XP  |
| 23 | Web Scraping with BeautifulSoup and Requests      | ■ Reading  | 35 min · 75 XP  |
| 24 | Introduction to NumPy and Pandas                  | ■ Reading  | 35 min · 75 XP  |
| 25 | Automation Practice: Web Scraper and Data Cleaner | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 26 | Data Visualization with Matplotlib                   | ■ Reading  | 35 min · 75 XP  |
| 27 | Working with Databases: SQLite and SQLAlchemy        | ■ Reading  | 35 min · 75 XP  |
| 28 | Testing with unittest and pytest                     | ■ Reading  | 35 min · 75 XP  |
| 29 | Concurrency: Threading, Multiprocessing, and asyncio | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Python Practice: Data Pipeline              | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                       |           |                |
|----|-------------------------------------------------------|-----------|----------------|
| 31 | Packaging and Distributing Python Projects            | ■ Reading | 30 min · 75 XP |
| 32 | Type Hints and Static Analysis with mypy              | ■ Reading | 35 min · 75 XP |
| 33 | Performance: Profiling and Optimization Techniques    | ■ Reading | 35 min · 75 XP |
| 34 | Python in Production: Logging, Config, and Deployment | ■ Reading | 35 min · 75 XP |



# Modern JavaScript

■ Pending

Slug: javascript · Difficulty: Intermediate · Duration: 32h planned · 22h lessons · Lessons: 35 (■28 ■5 ■0 ■2) · Tags: ES2024, Async, DOM, Modules, Closures

## Module 1

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 01 | JavaScript Fundamentals: Variables, Types, and Scope      | ■ Reading  | 30 min · 75 XP  |
| 02 | Functions: Declarations, Expressions, and Arrow Functions | ■ Reading  | 30 min · 75 XP  |
| 03 | Objects, Prototypes, and the Prototype Chain              | ■ Reading  | 35 min · 75 XP  |
| 04 | Arrays and Array Methods: Map, Filter, Reduce             | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: DOM Manipulation Challenges          | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 06 | Closures, Currying, and Higher-Order Functions       | ■ Reading  | 35 min · 75 XP  |
| 07 | The Event Loop: Call Stack, Web APIs, and Task Queue | ■ Reading  | 35 min · 75 XP  |
| 08 | Promises and the Microtask Queue                     | ■ Reading  | 35 min · 75 XP  |
| 09 | Async/Await and Error Handling Patterns              | ■ Reading  | 35 min · 75 XP  |
| 10 | Async Practice: Fetch API and Data Dashboard         | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 11 | ES Modules: Import, Export, and Module Bundlers | ■ Reading  | 35 min · 75 XP  |
| 12 | Classes and Object-Oriented JavaScript          | ■ Reading  | 35 min · 75 XP  |
| 13 | Iterators, Generators, and Symbols              | ■ Reading  | 35 min · 75 XP  |
| 14 | WeakMap, WeakSet, and Memory Management         | ■ Reading  | 35 min · 75 XP  |
| 15 | ES2024 Features Assessment and Practice         | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                        |           |                 |
|----|--------------------------------------------------------|-----------|-----------------|
| 16 | The DOM: Querying, Traversal, and Manipulation         | ■ Reading | 35 min · 75 XP  |
| 17 | Events: Bubbling, Delegation, and Custom Events        | ■ Reading | 35 min · 75 XP  |
| 18 | Web APIs: Fetch, LocalStorage, and IndexedDB           | ■ Reading | 35 min · 75 XP  |
| 19 | Web Workers and SharedArrayBuffer                      | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Interactive Data Visualization App | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 21 | Functional Programming Patterns in JavaScript  | ■ Reading  | 35 min · 75 XP  |
| 22 | Design Patterns: Module, Observer, and Factory | ■ Reading  | 35 min · 75 XP  |
| 23 | Error Handling, Debugging, and Source Maps     | ■ Reading  | 35 min · 75 XP  |
| 24 | Testing: Jest, Vitest, and Testing Library     | ■ Reading  | 35 min · 75 XP  |
| 25 | Patterns Practice: Design a Plugin System      | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 26 | TypeScript Fundamentals for JavaScript Developers | ■ Reading  | 35 min · 75 XP  |
| 27 | Performance Optimization: Rendering and Memory    | ■ Reading  | 35 min · 75 XP  |
| 28 | Security: XSS, CSRF, and Content Security Policy  | ■ Reading  | 35 min · 75 XP  |
| 29 | Node.js Basics for JavaScript Developers          | ■ Reading  | 35 min · 75 XP  |
| 30 | Performance and Security Hardening Exercise       | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                   |           |                |
|----|---------------------------------------------------|-----------|----------------|
| 31 | Tooling: ESLint, Prettier, and Build Pipelines    | ■ Reading | 30 min · 75 XP |
| 32 | Bundlers: Vite, Webpack, and esbuild              | ■ Reading | 35 min · 75 XP |
| 33 | JavaScript Runtime Internals: V8 and SpiderMonkey | ■ Reading | 35 min · 75 XP |
| 34 | Publishing Packages and Versioning with npm       | ■ Reading | 30 min · 75 XP |





# TypeScript Deep Dive

■ Pending

Slug: **typescript** · Difficulty: **Intermediate** · Duration: **24h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Types, Generics, Decorators, Utility Types

## Module 1

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 01 | TypeScript Fundamentals: Types, Interfaces, and Enums | ■ Reading  | 30 min · 75 XP  |
| 02 | Type Inference and Type Widening                      | ■ Reading  | 30 min · 75 XP  |
| 03 | Functions: Typed Parameters, Overloads, and Generics  | ■ Reading  | 35 min · 75 XP  |
| 04 | Classes with Access Modifiers and Decorators          | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Type-Safe Utility Library        | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 06 | Generics: Constraints, Defaults, and Conditional Types | ■ Reading  | 35 min · 75 XP  |
| 07 | Mapped Types and Template Literal Types                | ■ Reading  | 35 min · 75 XP  |
| 08 | Discriminated Unions and Exhaustive Checks             | ■ Reading  | 35 min · 75 XP  |
| 09 | Utility Types: Partial, Required, Pick, Omit, Record   | ■ Reading  | 35 min · 75 XP  |
| 10 | Advanced Types Practice: Type-Safe Event System        | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 11 | Module System: Declaration Files and DefinitelyTyped | ■ Reading  | 35 min · 75 XP  |
| 12 | Namespaces vs Modules: When to Use Each              | ■ Reading  | 30 min · 75 XP  |
| 13 | Declaration Merging and Ambient Declarations         | ■ Reading  | 35 min · 75 XP  |
| 14 | The TypeScript Compiler: tsconfig.json Deep Dive     | ■ Reading  | 35 min · 75 XP  |
| 15 | Module and Configuration Practice                    | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                      |           |                 |
|----|------------------------------------------------------|-----------|-----------------|
| 16 | Type Guards and Narrowing Patterns                   | ■ Reading | 35 min · 75 XP  |
| 17 | Infer Keyword and Type Inference in Generics         | ■ Reading | 35 min · 75 XP  |
| 18 | Recursive Types and Deeply Nested Structures         | ■ Reading | 35 min · 75 XP  |
| 19 | Variance: Covariance, Contravariance, and Invariance | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Type-Safe REST API Client        | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | TypeScript with React: Props, State, and Hooks Typing | ■ Reading  | 35 min · 75 XP  |
| 22 | TypeScript with Node.js and Express                   | ■ Reading  | 35 min · 75 XP  |
| 23 | Testing TypeScript: ts-jest and Type-Safe Mocks       | ■ Reading  | 35 min · 75 XP  |
| 24 | Performance: Type Instantiation and Compilation Speed | ■ Reading  | 35 min · 75 XP  |
| 25 | Framework Integration Practice                        | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | Design Patterns in TypeScript: Repository, CQRS, DI | ■ Reading  | 35 min · 75 XP  |
| 27 | Strict Mode and Gradual Typing Migration            | ■ Reading  | 35 min · 75 XP  |
| 28 | Decorators and Metadata Reflection                  | ■ Reading  | 35 min · 75 XP  |
| 29 | TypeScript 5.x New Features and Roadmap             | ■ Reading  | 35 min · 75 XP  |
| 30 | Production Patterns Practice                        | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                  |           |                |
|----|--------------------------------------------------|-----------|----------------|
| 31 | Monorepo TypeScript Setup with Turborepo         | ■ Reading | 35 min · 75 XP |
| 32 | Code Generation from Types: Zod and Prisma       | ■ Reading | 35 min · 75 XP |
| 33 | Linting and Formatting: ESLint with TypeScript   | ■ Reading | 30 min · 75 XP |
| 34 | Publishing Typed Libraries and Declaration Files | ■ Reading | 35 min · 75 XP |



## Go Essentials

■ Pending

Slug: **go** · Difficulty: **Intermediate** · Duration: **30h planned · 22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Goroutines, Channels, Interfaces, Microservices

### Module 1

|    |                                                            |            |                 |
|----|------------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Go: Philosophy, Toolchain, and Hello World | ■ Reading  | 30 min · 75 XP  |
| 02 | Variables, Types, and Zero Values                          | ■ Reading  | 30 min · 75 XP  |
| 03 | Functions, Multiple Returns, and Named Results             | ■ Reading  | 30 min · 75 XP  |
| 04 | Control Structures: For, If, Switch, and Defer             | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: CLI Tools in Go                       | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 06 | Structs, Methods, and Embedded Types                | ■ Reading  | 35 min · 75 XP  |
| 07 | Interfaces: Implicit Implementation and Composition | ■ Reading  | 35 min · 75 XP  |
| 08 | Pointers: Value vs Reference Semantics              | ■ Reading  | 35 min · 75 XP  |
| 09 | Error Handling: Idiomatic Go Error Patterns         | ■ Reading  | 35 min · 75 XP  |
| 10 | OOP Practice: Building a File System Abstraction    | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 11 | Slices, Maps, and Arrays in Depth           | ■ Reading  | 35 min · 75 XP  |
| 12 | Goroutines and the Go Scheduler             | ■ Reading  | 35 min · 75 XP  |
| 13 | Channels: Communication and Synchronization | ■ Reading  | 35 min · 75 XP  |
| 14 | Select Statement and Timeout Patterns       | ■ Reading  | 35 min · 75 XP  |
| 15 | Concurrency Practice: Web Crawler           | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                      |           |                 |
|----|------------------------------------------------------|-----------|-----------------|
| 16 | sync Package: Mutex, WaitGroup, and Once             | ■ Reading | 35 min · 75 XP  |
| 17 | Context: Cancellation and Deadlines                  | ■ Reading | 35 min · 75 XP  |
| 18 | Go Modules and Dependency Management                 | ■ Reading | 30 min · 75 XP  |
| 19 | Testing: go test, Table-Driven Tests, and Benchmarks | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Concurrent REST API Server       | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 21 | Standard Library: net/http, encoding/json, os | ■ Reading  | 35 min · 75 XP  |
| 22 | Building REST APIs with net/http and Chi      | ■ Reading  | 35 min · 75 XP  |
| 23 | Database Access with database/sql and sqlx    | ■ Reading  | 35 min · 75 XP  |
| 24 | Middleware Patterns and Request Lifecycle     | ■ Reading  | 35 min · 75 XP  |
| 25 | API Development Practice: CRUD Service        | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 26 | Reflection and the reflect Package             | ■ Reading  | 35 min · 75 XP  |
| 27 | Code Generation with go generate               | ■ Reading  | 35 min · 75 XP  |
| 28 | Profiling and Optimizing Go Applications       | ■ Reading  | 35 min · 75 XP  |
| 29 | Memory Management and Garbage Collection in Go | ■ Reading  | 35 min · 75 XP  |
| 30 | Performance Optimization Practice              | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                                   |           |                |
|----|---------------------------------------------------|-----------|----------------|
| 31 | Building CLI Tools with cobra                     | ■ Reading | 30 min · 75 XP |
| 32 | gRPC and Protocol Buffers in Go                   | ■ Reading | 35 min · 75 XP |
| 33 | Containerizing Go Applications with Docker        | ■ Reading | 30 min · 75 XP |
| 34 | Deployment and Observability: Metrics and Tracing | ■ Reading | 35 min · 75 XP |



# Rust Systems Programming

■ Pending

Slug: `rust` · Difficulty: **Advanced** · Duration: 50h planned · 22h lessons · Lessons: 35 (■28 ■5 ■0 ■2) · Tags: Ownership, Borrow Checker, Concurrency, WASM

## Module 1

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 01 | Introduction to Rust: Safety, Speed, and Ownership | ■ Reading  | 30 min · 75 XP  |
| 02 | Variables, Data Types, and Mutability              | ■ Reading  | 30 min · 75 XP  |
| 03 | Ownership: Move, Copy, and Drop                    | ■ Reading  | 35 min · 75 XP  |
| 04 | Borrowing and the Borrow Checker                   | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Ownership Challenges          | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | Lifetimes: Annotating and Elision Rules           | ■ Reading  | 35 min · 75 XP  |
| 07 | Structs, Impl Blocks, and Methods                 | ■ Reading  | 35 min · 75 XP  |
| 08 | Enums, Option, and Result                         | ■ Reading  | 35 min · 75 XP  |
| 09 | Pattern Matching with match and if let            | ■ Reading  | 35 min · 75 XP  |
| 10 | Type System Practice: Building a Shape Calculator | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 11 | Traits: Defining and Implementing Behaviour | ■ Reading  | 35 min · 75 XP  |
| 12 | Generics: Type Parameters and Trait Bounds  | ■ Reading  | 35 min · 75 XP  |
| 13 | Closures and the Fn, FnMut, FnOnce Traits   | ■ Reading  | 35 min · 75 XP  |
| 14 | Iterators and the Iterator Trait            | ■ Reading  | 35 min · 75 XP  |
| 15 | Traits and Generics Practice                | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 16 | Vectors, Strings, and Hash Maps                   | ■ Reading | 35 min · 75 XP  |
| 17 | Error Handling: unwrap, ?, and Custom Error Types | ■ Reading | 35 min · 75 XP  |
| 18 | Modules, Crates, and Cargo                        | ■ Reading | 30 min · 75 XP  |
| 19 | Testing in Rust: Unit, Integration, and Doc Tests | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Command-Line File Utility     | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 21 | Smart Pointers: Box, Rc, RefCell, and Arc   | ■ Reading  | 35 min · 75 XP  |
| 22 | Concurrency: Threads, Mutex, and Channels   | ■ Reading  | 35 min · 75 XP  |
| 23 | Async Rust: Futures, async/await, and Tokio | ■ Reading  | 35 min · 75 XP  |
| 24 | Unsafe Rust: When and How to Use It         | ■ Reading  | 35 min · 75 XP  |
| 25 | Concurrency Practice: Async Web Client      | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                      |            |                 |
|----|--------------------------------------|------------|-----------------|
| 26 | Macros: Declarative and Procedural   | ■ Reading  | 35 min · 75 XP  |
| 27 | The Foreign Function Interface (FFI) | ■ Reading  | 35 min · 75 XP  |
| 28 | WebAssembly with wasm-bindgen        | ■ Reading  | 35 min · 75 XP  |
| 29 | Building Web APIs with Axum          | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Rust Features Practice      | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                |           |                |
|----|------------------------------------------------|-----------|----------------|
| 31 | Performance Profiling with perf and flamegraph | ■ Reading | 35 min · 75 XP |
| 32 | Memory Layout, Alignment, and SIMD             | ■ Reading | 35 min · 75 XP |
| 33 | Rust in Embedded Systems                       | ■ Reading | 35 min · 75 XP |
| 34 | Publishing to crates.io and Documentation      | ■ Reading | 30 min · 75 XP |



# Frontend Development

## React 19 & Ecosystem

■ Pending

Slug: **react** · Difficulty: **Intermediate** · Duration: **36h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Hooks, Context, Suspense, RSC, TanStack

### Module 1

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 01 | Introduction to React 19 and the Component Model   | ■ Reading  | 30 min · 75 XP  |
| 02 | JSX, Rendering, and the Virtual DOM                | ■ Reading  | 30 min · 75 XP  |
| 03 | Props, State, and the Component Lifecycle          | ■ Reading  | 35 min · 75 XP  |
| 04 | Event Handling and Synthetic Events                | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Interactive Component Library | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 06 | useState and useReducer: Managing Local State | ■ Reading  | 35 min · 75 XP  |
| 07 | useEffect: Side Effects and Cleanup           | ■ Reading  | 35 min · 75 XP  |
| 08 | useRef, useMemo, and useCallback              | ■ Reading  | 35 min · 75 XP  |
| 09 | Custom Hooks: Extracting and Reusing Logic    | ■ Reading  | 35 min · 75 XP  |
| 10 | Hooks Practice: Build a Real-Time Dashboard   | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 11 | React 19 Actions and the useActionState Hook     | ■ Reading  | 35 min · 75 XP  |
| 12 | Server Components and the Client/Server Boundary | ■ Reading  | 35 min · 75 XP  |
| 13 | Suspense, Transitions, and useOptimistic         | ■ Reading  | 35 min · 75 XP  |
| 14 | Context API: Global State Without Libraries      | ■ Reading  | 35 min · 75 XP  |
| 15 | React 19 Features Assessment                     | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                      |           |                 |
|----|------------------------------------------------------|-----------|-----------------|
| 16 | React Router v7: Client-Side Navigation              | ■ Reading | 35 min · 75 XP  |
| 17 | Data Fetching Patterns: SWR and TanStack Query       | ■ Reading | 35 min · 75 XP  |
| 18 | Forms: Controlled, Uncontrolled, and React Hook Form | ■ Reading | 35 min · 75 XP  |
| 19 | Accessibility in React: ARIA and Focus Management    | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Full-Featured Blog Application   | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 21 | State Management at Scale with Zustand                  | ■ Reading  | 35 min · 75 XP  |
| 22 | Performance: Profiling, Memoization, and Code Splitting | ■ Reading  | 35 min · 75 XP  |
| 23 | Testing React: Vitest and React Testing Library         | ■ Reading  | 35 min · 75 XP  |
| 24 | Styling: CSS Modules, Tailwind, and CSS-in-JS           | ■ Reading  | 35 min · 75 XP  |
| 25 | Performance and Testing Practice                        | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | Animation with Framer Motion                        | ■ Reading  | 35 min · 75 XP  |
| 27 | React and TypeScript: Typing Props, Hooks, and Refs | ■ Reading  | 35 min · 75 XP  |
| 28 | Error Boundaries and Error Handling Patterns        | ■ Reading  | 35 min · 75 XP  |
| 29 | Micro-Frontends and Module Federation               | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Patterns Practice                          | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                                |           |                |
|----|------------------------------------------------|-----------|----------------|
| 31 | Deployment: Vercel, Netlify, and Static Export | ■ Reading | 30 min · 75 XP |
|----|------------------------------------------------|-----------|----------------|



|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 32 | Monitoring: React DevTools and Error Tracking     | ■ Reading | 30 min · 75 XP  |
| 33 | Security: XSS Prevention and Secure Data Handling | ■ Reading | 35 min · 75 XP  |
| 34 | React Native Overview for React Developers        | ■ Reading | 30 min · 75 XP  |
| 35 | Capstone: Production-Grade React Application      | ■ Project | 60 min · 150 XP |

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# Next.js App Router

■ Pending

Slug: [nextjs](#) · Difficulty: **Intermediate** · Duration: **30h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: App Router, RSC, ISR, Server Actions

## Module 1

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 01 | Introduction to Next.js and the App Router        | ■ Reading  | 30 min · 75 XP  |
| 02 | File-Based Routing: Pages, Layouts, and Templates | ■ Reading  | 30 min · 75 XP  |
| 03 | Server Components vs Client Components            | ■ Reading  | 35 min · 75 XP  |
| 04 | Fetching Data in Server Components                | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Multi-Page Marketing Site    | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 06 | Static Generation and Dynamic Rendering                 | ■ Reading  | 35 min · 75 XP  |
| 07 | Caching: Full Route Cache, Data Cache, and Revalidation | ■ Reading  | 35 min · 75 XP  |
| 08 | Incremental Static Regeneration (ISR)                   | ■ Reading  | 35 min · 75 XP  |
| 09 | Route Handlers and Building API Endpoints               | ■ Reading  | 35 min · 75 XP  |
| 10 | Caching and Rendering Practice                          | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                              |            |                 |
|----|----------------------------------------------|------------|-----------------|
| 11 | Server Actions: Mutations Without API Routes | ■ Reading  | 35 min · 75 XP  |
| 12 | Middleware: Auth, Redirects, and A/B Testing | ■ Reading  | 35 min · 75 XP  |
| 13 | Parallel and Intercepting Routes             | ■ Reading  | 35 min · 75 XP  |
| 14 | Loading UI, Suspense, and Streaming          | ■ Reading  | 35 min · 75 XP  |
| 15 | Server Actions Assessment: Form-Driven App   | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                 |           |                 |
|----|-------------------------------------------------|-----------|-----------------|
| 16 | Authentication with NextAuth.js v5              | ■ Reading | 35 min · 75 XP  |
| 17 | Database Integration: Prisma and Drizzle ORM    | ■ Reading | 35 min · 75 XP  |
| 18 | Optimistic Updates and useOptimistic            | ■ Reading | 35 min · 75 XP  |
| 19 | Image, Font, and Script Optimisation            | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Full-Stack E-Commerce Store | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 21 | Internationalisation (i18n) in Next.js          | ■ Reading  | 35 min · 75 XP  |
| 22 | Testing: Playwright, Vitest, and MSW            | ■ Reading  | 35 min · 75 XP  |
| 23 | TypeScript Integration and Type-Safe APIs       | ■ Reading  | 35 min · 75 XP  |
| 24 | Monitoring and Observability with OpenTelemetry | ■ Reading  | 35 min · 75 XP  |
| 25 | Testing and Monitoring Practice                 | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 26 | Deployment to Vercel: Edge Functions and Preview   | ■ Reading  | 30 min · 75 XP  |
| 27 | Self-Hosting Next.js on Node.js and Docker         | ■ Reading  | 35 min · 75 XP  |
| 28 | Performance: Core Web Vitals and Lighthouse        | ■ Reading  | 35 min · 75 XP  |
| 29 | Security: CSP, Rate Limiting, and Input Validation | ■ Reading  | 35 min · 75 XP  |
| 30 | Deployment and Performance Hardening               | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                    |           |                |
|----|----------------------------------------------------|-----------|----------------|
| 31 | Micro-Frontends with Next.js and Module Federation | ■ Reading | 35 min · 75 XP |
| 32 | Multi-Tenant Architecture in Next.js               | ■ Reading | 35 min · 75 XP |
| 33 | CMS Integration: Contentful, Sanity, and MDX       | ■ Reading | 35 min · 75 XP |
| 34 | Monorepo Setup with Turborepo and Next.js          | ■ Reading | 35 min · 75 XP |



# Angular 18 Enterprise

■ Pending

Slug: **angular** · Difficulty: **Intermediate** · Duration: **42h planned · 22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Signals, Standalone, RxJS, NgRx

## Module 1

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Angular 18 and the CLI               | ■ Reading  | 30 min · 75 XP  |
| 02 | Components, Templates, and Data Binding              | ■ Reading  | 30 min · 75 XP  |
| 03 | Standalone Components and the New Bootstrap API      | ■ Reading  | 35 min · 75 XP  |
| 04 | Directives: Built-In and Custom Attribute Directives | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Component-Based Dashboard       | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 06 | Signals: The New Reactivity Primitive                   | ■ Reading  | 35 min · 75 XP  |
| 07 | computed() and effect(): Derived State and Side Effects | ■ Reading  | 35 min · 75 XP  |
| 08 | Signal-Based Inputs and Outputs                         | ■ Reading  | 35 min · 75 XP  |
| 09 | Control Flow: @if, @for, and @switch Syntax             | ■ Reading  | 35 min · 75 XP  |
| 10 | Signals Practice: Reactive Shopping Cart                | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 11 | Dependency Injection: Providers and Hierarchical DI     | ■ Reading  | 35 min · 75 XP  |
| 12 | Services and the inject() Function                      | ■ Reading  | 35 min · 75 XP  |
| 13 | HTTP Client: Requests, Interceptors, and Error Handling | ■ Reading  | 35 min · 75 XP  |
| 14 | Angular Router: Lazy Loading and Route Guards           | ■ Reading  | 35 min · 75 XP  |
| 15 | DI and Services Assessment                              | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                             |           |                 |
|----|---------------------------------------------|-----------|-----------------|
| 16 | Reactive Forms and Form Validation          | ■ Reading | 35 min · 75 XP  |
| 17 | Angular Animations and Transition API       | ■ Reading | 35 min · 75 XP  |
| 18 | Pipes: Built-In and Custom Pipe Development | ■ Reading | 35 min · 75 XP  |
| 19 | NgRx Signals Store: State Management        | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Enterprise HR Portal    | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | RxJS in Angular: Observables, Operators, and Subjects | ■ Reading  | 35 min · 75 XP  |
| 22 | Change Detection: OnPush and Zoneless Angular         | ■ Reading  | 35 min · 75 XP  |
| 23 | Testing Angular: Karma, Jest, and Spectator           | ■ Reading  | 35 min · 75 XP  |
| 24 | Internationalization (i18n) and Localisation          | ■ Reading  | 35 min · 75 XP  |
| 25 | RxJS and Testing Practice                             | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | Angular CDK: Overlay, Drag-and-Drop, Virtual Scroll | ■ Reading  | 35 min · 75 XP  |
| 27 | Server-Side Rendering with Angular Universal        | ■ Reading  | 35 min · 75 XP  |
| 28 | Progressive Web Apps with Angular                   | ■ Reading  | 35 min · 75 XP  |
| 29 | Micro-Frontend Architecture with Module Federation  | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Angular Patterns Practice                  | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                        |           |                |
|----|--------------------------------------------------------|-----------|----------------|
| 31 | Performance: Bundle Optimization and Tree Shaking      | ■ Reading | 35 min · 75 XP |
| 32 | Security: CSRF Protection and Content Security         | ■ Reading | 35 min · 75 XP |
| 33 | CI/CD for Angular: GitHub Actions and Firebase Hosting | ■ Reading | 30 min · 75 XP |
| 34 | Angular DevTools and Profiling Applications            | ■ Reading | 30 min · 75 XP |



# Vue.js 3 Composition API

■ Pending

Slug: [vuejs](#) · Difficulty: **Beginner** · Duration: **28h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Composition API, Pinia, Nuxt 3, TypeScript

## Module 1

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 01 | Introduction to Vue 3 and the Composition API | ■ Reading  | 30 min · 75 XP  |
| 02 | Template Syntax, Directives, and Data Binding | ■ Reading  | 30 min · 75 XP  |
| 03 | Reactivity Fundamentals: ref() and reactive() | ■ Reading  | 35 min · 75 XP  |
| 04 | Computed Properties and Watchers              | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Interactive Todo App     | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 06 | Components: Props, Emits, and Slots              | ■ Reading  | 35 min · 75 XP  |
| 07 | Component Lifecycle Hooks in Composition API     | ■ Reading  | 35 min · 75 XP  |
| 08 | Composables: Reusable Logic with Composition API | ■ Reading  | 35 min · 75 XP  |
| 09 | Provide and Inject: Dependency Injection in Vue  | ■ Reading  | 35 min · 75 XP  |
| 10 | Components Practice: Reusable UI Library         | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 11 | Vue Router 4: Navigation, Guards, and Lazy Loading | ■ Reading  | 35 min · 75 XP  |
| 12 | Pinia: State Management for Vue 3                  | ■ Reading  | 35 min · 75 XP  |
| 13 | Forms: v-model, Validation, and VeeValidate        | ■ Reading  | 35 min · 75 XP  |
| 14 | Transitions and Animations in Vue                  | ■ Reading  | 35 min · 75 XP  |
| 15 | Router and State Management Assessment             | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                         |           |                 |
|----|---------------------------------------------------------|-----------|-----------------|
| 16 | Async Components and Suspense                           | ■ Reading | 35 min · 75 XP  |
| 17 | Custom Directives and Plugins                           | ■ Reading | 35 min · 75 XP  |
| 18 | Renderless Components and the Headless Pattern          | ■ Reading | 35 min · 75 XP  |
| 19 | TypeScript with Vue 3: Typed Components and Composables | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Full-Featured E-Commerce SPA        | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | Nuxt 3: File-Based Routing and Server-Side Rendering  | ■ Reading  | 35 min · 75 XP  |
| 22 | Nuxt 3: Data Fetching with useFetch and useAsyncData  | ■ Reading  | 35 min · 75 XP  |
| 23 | Nuxt 3: API Routes and Server Middleware              | ■ Reading  | 35 min · 75 XP  |
| 24 | Testing Vue Components with Vitest and Vue Test Utils | ■ Reading  | 35 min · 75 XP  |
| 25 | Nuxt 3 and Testing Practice                           | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 26 | Performance: Virtual Scrolling, Lazy Loading, Memoization | ■ Reading  | 35 min · 75 XP  |
| 27 | Accessibility in Vue Applications                         | ■ Reading  | 30 min · 75 XP  |
| 28 | Internationalization with vue-i18n                        | ■ Reading  | 30 min · 75 XP  |
| 29 | Micro-Frontends with Vue                                  | ■ Reading  | 35 min · 75 XP  |
| 30 | Performance and Accessibility Practice                    | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                              |           |                |
|----|----------------------------------------------|-----------|----------------|
| 31 | Deployment: Netlify, Vercel, and Self-Hosted | ■ Reading | 30 min · 75 XP |
| 32 | Vite: Build Tool Deep Dive for Vue Projects  | ■ Reading | 35 min · 75 XP |
| 33 | Security Best Practices in Vue Applications  | ■ Reading | 35 min · 75 XP |
| 34 | Vue DevTools and Debugging Techniques        | ■ Reading | 30 min · 75 XP |



# Tailwind CSS Mastery

■ Pending

Slug: [tailwindcss](#) · Difficulty: **Beginner** · Duration: **16h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Utility-First, Design Systems, Animations

## Module 1

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Utility-First CSS and Tailwind Philosophy | ■ Reading  | 30 min · 75 XP  |
| 02 | Core Utilities: Spacing, Sizing, and Layout               | ■ Reading  | 30 min · 75 XP  |
| 03 | Typography: Font, Text, and List Utilities                | ■ Reading  | 30 min · 75 XP  |
| 04 | Colors, Backgrounds, and Gradients                        | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Landing Page with Tailwind           | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 06 | Flexbox Utilities: Alignment, Direction, and Wrapping | ■ Reading  | 35 min · 75 XP  |
| 07 | CSS Grid with Tailwind: Columns, Rows, and Placement  | ■ Reading  | 35 min · 75 XP  |
| 08 | Responsive Design: Breakpoints and Mobile-First       | ■ Reading  | 35 min · 75 XP  |
| 09 | Dark Mode: Class Strategy and Media Strategy          | ■ Reading  | 35 min · 75 XP  |
| 10 | Layout Practice: Responsive Dashboard                 | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 11 | Hover, Focus, and Other State Variants               | ■ Reading  | 35 min · 75 XP  |
| 12 | Transitions, Animations, and Motion                  | ■ Reading  | 35 min · 75 XP  |
| 13 | Borders, Shadows, Rings, and Outlines                | ■ Reading  | 30 min · 75 XP  |
| 14 | Tailwind Configuration: tailwind.config.js Deep Dive | ■ Reading  | 35 min · 75 XP  |
| 15 | Interactivity and Animation Practice                 | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                       |           |                 |
|----|-------------------------------------------------------|-----------|-----------------|
| 16 | Custom Design Tokens: Extending the Theme             | ■ Reading | 35 min · 75 XP  |
| 17 | Arbitrary Values and CSS Variables                    | ■ Reading | 35 min · 75 XP  |
| 18 | @apply Directive and Managing Component Styles        | ■ Reading | 35 min · 75 XP  |
| 19 | Tailwind Plugins: Typography, Forms, and Aspect Ratio | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Design System with Tailwind       | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 21 | Component Patterns: Card, Button, Modal, and Navbar | ■ Reading  | 35 min · 75 XP  |
| 22 | Building Accessible Components with Tailwind        | ■ Reading  | 35 min · 75 XP  |
| 23 | Tailwind with React: clsx and tailwind-merge        | ■ Reading  | 35 min · 75 XP  |
| 24 | Tailwind with Vue and Angular                       | ■ Reading  | 30 min · 75 XP  |
| 25 | Component Library Practice                          | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | Headless UI: Accessible Component Primitives        | ■ Reading  | 35 min · 75 XP  |
| 27 | Radix UI and shadcn/ui with Tailwind                | ■ Reading  | 35 min · 75 XP  |
| 28 | Optimizing Bundle Size: PurgeCSS and Content Config | ■ Reading  | 35 min · 75 XP  |
| 29 | Tailwind CSS v4: What Changed and Migration Guide   | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Tailwind and Framework Integration         | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                  |           |                |
|----|--------------------------------------------------|-----------|----------------|
| 31 | Building a Custom Component Library for Teams    | ■ Reading | 35 min · 75 XP |
| 32 | Storybook Integration with Tailwind              | ■ Reading | 30 min · 75 XP |
| 33 | Design Tokens: Bridging Figma and Tailwind       | ■ Reading | 35 min · 75 XP |
| 34 | Performance: Critical CSS and Loading Strategies | ■ Reading | 30 min · 75 XP |





# Backend Development

## Node.js & Express

■ Pending

Slug: **nodejs** · Difficulty: **Intermediate** · Duration: **32h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: REST, Express, Fastify, Event Loop, Streams

### Module 1

|    |                                                          |            |                 |
|----|----------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Node.js: Event Loop and Non-Blocking I/O | ■ Reading  | 30 min · 75 XP  |
| 02 | Modules: CommonJS, ESM, and the Module System            | ■ Reading  | 30 min · 75 XP  |
| 03 | The Core APIs: fs, path, os, and process                 | ■ Reading  | 35 min · 75 XP  |
| 04 | Streams and Buffers: Efficient Data Processing           | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: File Processing CLI Tool            | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 06 | HTTP Module: Building a Server from Scratch            | ■ Reading  | 35 min · 75 XP  |
| 07 | Express.js: Routing, Middleware, and Request Lifecycle | ■ Reading  | 35 min · 75 XP  |
| 08 | REST API Design: Resources, Verbs, and Status Codes    | ■ Reading  | 35 min · 75 XP  |
| 09 | Middleware Patterns: Auth, Logging, and Error Handling | ■ Reading  | 35 min · 75 XP  |
| 10 | Express Practice: Full CRUD REST API                   | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 11 | Fastify: High-Performance Alternative to Express     | ■ Reading  | 35 min · 75 XP  |
| 12 | Validation with Zod and Input Sanitization           | ■ Reading  | 35 min · 75 XP  |
| 13 | Authentication: JWT, Cookies, and Session Management | ■ Reading  | 35 min · 75 XP  |
| 14 | Database Integration: PostgreSQL with node-postgres  | ■ Reading  | 35 min · 75 XP  |
| 15 | Authentication and Database Practice                 | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                  |           |                 |
|----|--------------------------------------------------|-----------|-----------------|
| 16 | ORM with Prisma: Schema, Migrations, and Queries | ■ Reading | 35 min · 75 XP  |
| 17 | Caching with Redis: Strategies and TTL           | ■ Reading | 35 min · 75 XP  |
| 18 | WebSockets with ws and Socket.io                 | ■ Reading | 35 min · 75 XP  |
| 19 | Job Queues with BullMQ and Redis                 | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Real-Time Chat API           | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | Testing Node.js: Jest, Supertest, and Test Containers | ■ Reading  | 35 min · 75 XP  |
| 22 | Error Handling Patterns and Graceful Shutdown         | ■ Reading  | 35 min · 75 XP  |
| 23 | Logging with Winston and Structured Logging           | ■ Reading  | 35 min · 75 XP  |
| 24 | Environment Configuration and Secret Management       | ■ Reading  | 30 min · 75 XP  |
| 25 | Testing and Observability Practice                    | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | Event-Driven Architecture with EventEmitter         | ■ Reading  | 35 min · 75 XP  |
| 27 | File Uploads: Multipart, S3, and Streaming          | ■ Reading  | 35 min · 75 XP  |
| 28 | Rate Limiting, Throttling, and Security Headers     | ■ Reading  | 35 min · 75 XP  |
| 29 | Performance Profiling: node --inspect and Clinic.js | ■ Reading  | 35 min · 75 XP  |
| 30 | Performance and Security Hardening                  | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                             |           |                |
|----|---------------------------------------------|-----------|----------------|
| 31 | Clustering and Worker Threads for CPU Tasks | ■ Reading | 35 min · 75 XP |
|----|---------------------------------------------|-----------|----------------|

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 32 | Containerizing Node.js with Docker                | ■ Reading | 30 min · 75 XP  |
| 33 | Deployment: PM2, Health Checks, and Zero-Downtime | ■ Reading | 35 min · 75 XP  |
| 34 | Monitoring with OpenTelemetry and Prometheus      | ■ Reading | 35 min · 75 XP  |
| 35 | Capstone: Production-Grade Node.js API Platform   | ■ Project | 60 min · 150 XP |

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# Spring Boot 3

■ Pending

Slug: **spring-boot** · Difficulty: **Intermediate** · Duration: **44h planned · 22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Spring MVC, Security, JPA, Microservices, Reactive

## Module 1

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 01 | Introduction to Spring Boot 3 and Spring Framework  | ■ Reading  | 30 min · 75 XP  |
| 02 | Spring IoC Container and Dependency Injection       | ■ Reading  | 35 min · 75 XP  |
| 03 | Spring MVC: Controllers, RequestMapping, and DTOs   | ■ Reading  | 35 min · 75 XP  |
| 04 | Building REST APIs with @RestController             | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: CRUD REST API with Spring Boot | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | Spring Data JPA: Entities, Repositories, and JPQL | ■ Reading  | 35 min · 75 XP  |
| 07 | Database Migrations with Flyway and Liquibase     | ■ Reading  | 35 min · 75 XP  |
| 08 | Spring Boot Validation with Bean Validation API   | ■ Reading  | 35 min · 75 XP  |
| 09 | Exception Handling with @ControllerAdvice         | ■ Reading  | 35 min · 75 XP  |
| 10 | JPA and Data Access Practice                      | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 11 | Spring Security 6: Authentication and Authorization | ■ Reading  | 35 min · 75 XP  |
| 12 | JWT Authentication and OAuth2 Resource Server       | ■ Reading  | 35 min · 75 XP  |
| 13 | Method-Level Security and Custom Access Rules       | ■ Reading  | 35 min · 75 XP  |
| 14 | Spring Boot Actuator: Health, Metrics, and Info     | ■ Reading  | 35 min · 75 XP  |
| 15 | Security Assessment: Secured Multi-Role API         | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 16 | Spring WebFlux: Reactive Programming with Reactor | ■ Reading | 35 min · 75 XP  |
| 17 | Spring Cache Abstraction with Redis               | ■ Reading | 35 min · 75 XP  |
| 18 | Spring Events and ApplicationEvent                | ■ Reading | 35 min · 75 XP  |
| 19 | Scheduling Tasks with @Scheduled                  | ■ Reading | 30 min · 75 XP  |
| 20 | Mid-Course Project: Multi-Tenant SaaS Backend     | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 21 | Spring Cloud: Config Server and Service Discovery | ■ Reading  | 35 min · 75 XP  |
| 22 | Spring Cloud Gateway: API Gateway Patterns        | ■ Reading  | 35 min · 75 XP  |
| 23 | Resilience4j: Circuit Breaker and Retry           | ■ Reading  | 35 min · 75 XP  |
| 24 | Spring Kafka: Event-Driven Microservices          | ■ Reading  | 35 min · 75 XP  |
| 25 | Microservices Practice: Order Processing System   | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                               |            |                 |
|----|---------------------------------------------------------------|------------|-----------------|
| 26 | Testing Spring Boot: MockMvc, @SpringBootTest, Testcontainers | ■ Reading  | 35 min · 75 XP  |
| 27 | OpenAPI Documentation with springdoc                          | ■ Reading  | 30 min · 75 XP  |
| 28 | GraalVM Native Image for Spring Boot                          | ■ Reading  | 35 min · 75 XP  |
| 29 | Performance Tuning: Connection Pools, JPA N+1, Caching        | ■ Reading  | 35 min · 75 XP  |
| 30 | Testing and Performance Practice                              | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                          |           |                |
|----|----------------------------------------------------------|-----------|----------------|
| 31 | Containerizing Spring Boot with Docker and Jib           | ■ Reading | 30 min · 75 XP |
| 32 | Kubernetes Deployment for Spring Boot                    | ■ Reading | 35 min · 75 XP |
| 33 | Observability: Micrometer, Zipkin, and Grafana           | ■ Reading | 35 min · 75 XP |
| 34 | Spring Authorization Server: Building an OAuth2 Provider | ■ Reading | 35 min · 75 XP |



# Django & DRF

■ Pending

Slug: **django** · Difficulty: **Intermediate** · Duration: **34h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: DRF, Authentication, ORM, Celery, Channels

## Module 1

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Django: MVT Pattern and Project Structure | ■ Reading  | 30 min · 75 XP  |
| 02 | Models: Fields, Relationships, and Migrations             | ■ Reading  | 35 min · 75 XP  |
| 03 | Django ORM: QuerySets, Lookups, and Aggregations          | ■ Reading  | 35 min · 75 XP  |
| 04 | Views: Function-Based vs Class-Based Views                | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Blog with Full CRUD                  | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | URL Configuration and Namespaces                  | ■ Reading  | 30 min · 75 XP  |
| 07 | Templates: Inheritance, Tags, and Filters         | ■ Reading  | 35 min · 75 XP  |
| 08 | Forms: ModelForms, Validation, and Widgets        | ■ Reading  | 35 min · 75 XP  |
| 09 | Django Admin: Customisation and Permissions       | ■ Reading  | 35 min · 75 XP  |
| 10 | Forms and Admin Practice: Event Management System | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 11 | Django REST Framework: Serializers and ViewSets | ■ Reading  | 35 min · 75 XP  |
| 12 | DRF Authentication: Token, Session, and JWT     | ■ Reading  | 35 min · 75 XP  |
| 13 | DRF Permissions, Throttling, and Filtering      | ■ Reading  | 35 min · 75 XP  |
| 14 | API Versioning and Pagination Strategies        | ■ Reading  | 35 min · 75 XP  |
| 15 | DRF Assessment: RESTful E-Commerce API          | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                         |           |                 |
|----|---------------------------------------------------------|-----------|-----------------|
| 16 | Django Authentication: Login, Registration, and Signals | ■ Reading | 35 min · 75 XP  |
| 17 | File Uploads: Local Storage and AWS S3                  | ■ Reading | 35 min · 75 XP  |
| 18 | Celery: Asynchronous Tasks and Periodic Jobs            | ■ Reading | 35 min · 75 XP  |
| 19 | Django Channels: WebSockets and Real-Time Features      | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Real-Time Collaboration Platform    | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 21 | Caching with Django and Redis                      | ■ Reading  | 35 min · 75 XP  |
| 22 | Django Testing: Unit, Integration, and Factory Boy | ■ Reading  | 35 min · 75 XP  |
| 23 | Custom Management Commands and Admin Actions       | ■ Reading  | 30 min · 75 XP  |
| 24 | Django Signals and Post-Save Hooks                 | ■ Reading  | 35 min · 75 XP  |
| 25 | Async and Caching Practice                         | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                                  |            |                 |
|----|------------------------------------------------------------------|------------|-----------------|
| 26 | Database Optimization: Indexes, select_related, prefetch_related | ■ Reading  | 35 min · 75 XP  |
| 27 | Multi-Tenancy Patterns in Django                                 | ■ Reading  | 35 min · 75 XP  |
| 28 | Security: CSRF, SQL Injection, and XSS Prevention                | ■ Reading  | 35 min · 75 XP  |
| 29 | Django with Docker: Development and Production Setup             | ■ Reading  | 30 min · 75 XP  |
| 30 | Optimization and Security Practice                               | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                      |           |                |
|----|------------------------------------------------------|-----------|----------------|
| 31 | Deployment: Gunicorn, Nginx, and SSL                 | ■ Reading | 35 min · 75 XP |
| 32 | Static Files and Media in Production with WhiteNoise | ■ Reading | 30 min · 75 XP |
| 33 | Monitoring: Sentry, Logging, and Health Checks       | ■ Reading | 35 min · 75 XP |
| 34 | Scaling Django: Horizontal Scaling and Read Replicas | ■ Reading | 35 min · 75 XP |



# FastAPI & Async Python

■ Pending

Slug: **fastapi** · Difficulty: **Intermediate** · Duration: **24h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Async, Pydantic v2, OpenAPI, Dependency Injection

## Module 1

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 01 | Introduction to FastAPI: Async Python and OpenAPI    | ■ Reading  | 30 min · 75 XP  |
| 02 | Path Operations: Routes, Methods, and Status Codes   | ■ Reading  | 30 min · 75 XP  |
| 03 | Pydantic v2: Data Validation and Settings Management | ■ Reading  | 35 min · 75 XP  |
| 04 | Request Body, Query Parameters, and Path Parameters  | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Product Catalogue API           | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | Dependency Injection with Depends()               | ■ Reading  | 35 min · 75 XP  |
| 07 | Authentication: OAuth2 Password Flow and JWT      | ■ Reading  | 35 min · 75 XP  |
| 08 | Middleware: CORS, Logging, and Request Timing     | ■ Reading  | 35 min · 75 XP  |
| 09 | Background Tasks and the BackgroundTasks Class    | ■ Reading  | 30 min · 75 XP  |
| 10 | Auth and Middleware Practice: User API with Roles | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 11 | SQLAlchemy 2 with FastAPI: Async Sessions       | ■ Reading  | 35 min · 75 XP  |
| 12 | Alembic Migrations with FastAPI                 | ■ Reading  | 30 min · 75 XP  |
| 13 | Repository Pattern and Service Layer            | ■ Reading  | 35 min · 75 XP  |
| 14 | File Uploads and Streaming Responses            | ■ Reading  | 35 min · 75 XP  |
| 15 | Database Integration Practice: Task Manager API | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 16 | WebSockets in FastAPI                             | ■ Reading | 35 min · 75 XP  |
| 17 | Server-Sent Events and Streaming                  | ■ Reading | 35 min · 75 XP  |
| 18 | Celery Integration for Background Jobs            | ■ Reading | 35 min · 75 XP  |
| 19 | Caching with Redis and FastAPI                    | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Real-Time Notification System | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 21 | Testing FastAPI: pytest, httpx, and TestClient          | ■ Reading  | 35 min · 75 XP  |
| 22 | Error Handling and Custom Exception Handlers            | ■ Reading  | 35 min · 75 XP  |
| 23 | API Versioning Strategies                               | ■ Reading  | 30 min · 75 XP  |
| 24 | OpenAPI Customisation: Tags, Descriptions, and Examples | ■ Reading  | 30 min · 75 XP  |
| 25 | Testing and Documentation Practice                      | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 26 | Rate Limiting with slowapi                    | ■ Reading  | 30 min · 75 XP  |
| 27 | Security: Input Sanitisation and OWASP Top 10 | ■ Reading  | 35 min · 75 XP  |
| 28 | Containerising FastAPI with Docker            | ■ Reading  | 30 min · 75 XP  |
| 29 | Deployment: Uvicorn, Gunicorn, and HTTPS      | ■ Reading  | 35 min · 75 XP  |
| 30 | Security and Deployment Practice              | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                               |           |                |
|----|-----------------------------------------------|-----------|----------------|
| 31 | Performance: Profiling and Async Optimisation | ■ Reading | 35 min · 75 XP |
| 32 | Monitoring with Prometheus and OpenTelemetry  | ■ Reading | 35 min · 75 XP |
| 33 | GraphQL with Strawberry and FastAPI           | ■ Reading | 35 min · 75 XP |
| 34 | gRPC with FastAPI and grpcio                  | ■ Reading | 35 min · 75 XP |





# Microservices Architecture

■ Pending

Slug: **microservices** · Difficulty: **Advanced** · Duration: **40h planned · 22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: **DDD, Event-Driven, Saga, CQRS, Service Mesh**

## Module 1

|    |                                                          |            |                 |
|----|----------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Microservices: Principles and Trade-offs | ■ Reading  | 30 min · 75 XP  |
| 02 | Domain-Driven Design: Bounded Contexts and Aggregates    | ■ Reading  | 35 min · 75 XP  |
| 03 | Service Decomposition: Patterns and Anti-Patterns        | ■ Reading  | 35 min · 75 XP  |
| 04 | API Gateway Pattern: Routing, Auth, and Rate Limiting    | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Decompose a Monolith                | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | Synchronous Communication: REST and gRPC          | ■ Reading  | 35 min · 75 XP  |
| 07 | Asynchronous Messaging with Kafka                 | ■ Reading  | 35 min · 75 XP  |
| 08 | Event-Driven Architecture and Event Sourcing      | ■ Reading  | 35 min · 75 XP  |
| 09 | The Saga Pattern: Distributed Transactions        | ■ Reading  | 35 min · 75 XP  |
| 10 | Communication Practice: Order Service with Events | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 11 | CQRS: Separating Commands and Queries         | ■ Reading  | 35 min · 75 XP  |
| 12 | Database per Service and Polyglot Persistence | ■ Reading  | 35 min · 75 XP  |
| 13 | Service Discovery: Consul and Kubernetes DNS  | ■ Reading  | 35 min · 75 XP  |
| 14 | Load Balancing and Client-Side Load Balancing | ■ Reading  | 35 min · 75 XP  |
| 15 | CQRS and Data Practice                        | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                           |           |                 |
|----|-----------------------------------------------------------|-----------|-----------------|
| 16 | Resilience Patterns: Circuit Breaker, Retry, and Bulkhead | ■ Reading | 35 min · 75 XP  |
| 17 | Distributed Tracing with Jaeger and OpenTelemetry         | ■ Reading | 35 min · 75 XP  |
| 18 | Centralised Logging with ELK Stack                        | ■ Reading | 35 min · 75 XP  |
| 19 | Health Checks and Readiness Probes                        | ■ Reading | 30 min · 75 XP  |
| 20 | Mid-Course Project: Microservices E-Commerce Platform     | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 21 | Service Mesh with Istio: mTLS and Traffic Management | ■ Reading  | 35 min · 75 XP  |
| 22 | API Versioning in Microservices                      | ■ Reading  | 35 min · 75 XP  |
| 23 | Security: mTLS, Zero-Trust, and Secret Management    | ■ Reading  | 35 min · 75 XP  |
| 24 | Testing Microservices: Contract Testing with Pact    | ■ Reading  | 35 min · 75 XP  |
| 25 | Service Mesh and Security Practice                   | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 26 | Containerisation and Docker Compose for Development    | ■ Reading  | 30 min · 75 XP  |
| 27 | Kubernetes for Microservices: Deployments and Services | ■ Reading  | 35 min · 75 XP  |
| 28 | Helm Charts for Microservice Deployment                | ■ Reading  | 35 min · 75 XP  |
| 29 | CI/CD Pipelines for Microservices                      | ■ Reading  | 35 min · 75 XP  |
| 30 | Deployment and CI/CD Practice                          | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                            |           |                |
|----|------------------------------------------------------------|-----------|----------------|
| 31 | Scaling Strategies: Horizontal, Vertical, and Auto-Scaling | ■ Reading | 35 min · 75 XP |
| 32 | Cost Optimisation in Microservices Infrastructure          | ■ Reading | 35 min · 75 XP |
| 33 | Migration from Monolith: Strangler Fig Pattern             | ■ Reading | 35 min · 75 XP |
| 34 | Governance: Service Ownership and API Standards            | ■ Reading | 30 min · 75 XP |



# Cloud Computing

## AWS Solutions Architect

■ Pending

Slug: [aws](#) · Difficulty: **Intermediate** · Duration: **50h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: EC2, S3, Lambda, RDS, VPC, CDK

### Module 1

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 01 | Introduction to AWS: Global Infrastructure and IAM | ■ Reading  | 30 min · 75 XP  |
| 02 | IAM: Users, Roles, Policies, and Best Practices    | ■ Reading  | 35 min · 75 XP  |
| 03 | EC2: Instances, AMIs, and Auto Scaling Groups      | ■ Reading  | 35 min · 75 XP  |
| 04 | VPC: Subnets, Route Tables, and Security Groups    | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Secure Multi-Tier VPC         | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | S3: Buckets, Policies, Versioning, and Lifecycle  | ■ Reading  | 35 min · 75 XP  |
| 07 | RDS and Aurora: Managed Relational Databases      | ■ Reading  | 35 min · 75 XP  |
| 08 | ElastiCache: Redis and Memcached on AWS           | ■ Reading  | 35 min · 75 XP  |
| 09 | DynamoDB: NoSQL at Scale with Single-Table Design | ■ Reading  | 35 min · 75 XP  |
| 10 | Storage and Database Practice: Data Lake Setup    | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 11 | Lambda: Serverless Functions and Triggers         | ■ Reading  | 35 min · 75 XP  |
| 12 | API Gateway: REST and HTTP APIs                   | ■ Reading  | 35 min · 75 XP  |
| 13 | SQS and SNS: Messaging and Pub/Sub                | ■ Reading  | 35 min · 75 XP  |
| 14 | EventBridge: Event-Driven Architectures on AWS    | ■ Reading  | 35 min · 75 XP  |
| 15 | Serverless Practice: Event-Driven Order Processor | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                  |           |                 |
|----|--------------------------------------------------|-----------|-----------------|
| 16 | ECS and Fargate: Containerised Workloads         | ■ Reading | 35 min · 75 XP  |
| 17 | EKS: Managed Kubernetes on AWS                   | ■ Reading | 35 min · 75 XP  |
| 18 | CloudFront: CDN, Edge Functions, and Caching     | ■ Reading | 35 min · 75 XP  |
| 19 | Route 53: DNS, Health Checks, and Failover       | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Serverless SaaS Architecture | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 21 | CloudFormation: Infrastructure as Code        | ■ Reading  | 35 min · 75 XP  |
| 22 | AWS CDK: Cloud Infrastructure with TypeScript | ■ Reading  | 35 min · 75 XP  |
| 23 | CloudWatch: Logs, Metrics, and Alarms         | ■ Reading  | 35 min · 75 XP  |
| 24 | X-Ray: Distributed Tracing on AWS             | ■ Reading  | 35 min · 75 XP  |
| 25 | IaC and Observability Practice                | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 26 | Secrets Manager and Parameter Store                     | ■ Reading  | 30 min · 75 XP  |
| 27 | AWS WAF, Shield, and Security Hub                       | ■ Reading  | 35 min · 75 XP  |
| 28 | Cost Optimisation: Reserved Instances and Savings Plans | ■ Reading  | 35 min · 75 XP  |
| 29 | Multi-Region Architecture and Disaster Recovery         | ■ Reading  | 35 min · 75 XP  |
| 30 | Security and Cost Optimisation Practice                 | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                              |           |                |
|----|----------------------------------------------|-----------|----------------|
| 31 | AWS Well-Architected Framework: Five Pillars | ■ Reading | 35 min · 75 XP |
|----|----------------------------------------------|-----------|----------------|

|    |                                                    |           |                 |
|----|----------------------------------------------------|-----------|-----------------|
| 32 | Step Functions: Orchestrating Serverless Workflows | ■ Reading | 35 min · 75 XP  |
| 33 | Glue and Athena: Serverless Data Analytics         | ■ Reading | 35 min · 75 XP  |
| 34 | Exam Preparation: Solutions Architect Associate    | ■ Reading | 35 min · 75 XP  |
| 35 | Capstone: Resilient Multi-Region AWS Architecture  | ■ Project | 60 min · 150 XP |

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# DevOps & Infrastructure

## Docker & Containerization

■ Pending

Slug: **docker** · Difficulty: **Beginner** · Duration: **18h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Dockerfile, Compose, Networking, Security

### Module 1

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Containers and Docker Architecture   | ■ Reading  | 30 min · 75 XP  |
| 02 | Installing Docker and Running Your First Container   | ■ Reading  | 20 min · 50 XP  |
| 03 | Docker Images: Layers, Registries, and Pulling       | ■ Reading  | 30 min · 75 XP  |
| 04 | Writing Dockerfiles: Instructions and Best Practices | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Containerise a Web Application  | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 06 | Multi-Stage Builds for Smaller Production Images   | ■ Reading  | 35 min · 75 XP  |
| 07 | Docker Volumes: Persistent Data and Bind Mounts    | ■ Reading  | 35 min · 75 XP  |
| 08 | Docker Networking: Bridge, Host, and Overlay       | ■ Reading  | 35 min · 75 XP  |
| 09 | Environment Variables and Docker Secrets           | ■ Reading  | 30 min · 75 XP  |
| 10 | Intermediate Practice: Multi-Container Application | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 11 | Docker Compose: Defining Multi-Service Applications | ■ Reading  | 35 min · 75 XP  |
| 12 | Docker Compose: Networks, Volumes, and Profiles     | ■ Reading  | 35 min · 75 XP  |
| 13 | Health Checks and Dependency Management             | ■ Reading  | 30 min · 75 XP  |
| 14 | Docker Compose for Development Environments         | ■ Reading  | 35 min · 75 XP  |
| 15 | Docker Compose Practice: Full-Stack Dev Environment | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                        |           |                 |
|----|--------------------------------------------------------|-----------|-----------------|
| 16 | Container Security: Rootless Docker and Image Scanning | ■ Reading | 35 min · 75 XP  |
| 17 | Docker Content Trust and Image Signing                 | ■ Reading | 35 min · 75 XP  |
| 18 | Resource Limits: CPU, Memory, and cgroups              | ■ Reading | 35 min · 75 XP  |
| 19 | Logging Drivers and Centralised Log Aggregation        | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Production-Ready Container Stack   | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 21 | Docker Registry: Hosting a Private Registry | ■ Reading  | 35 min · 75 XP  |
| 22 | Pushing to Docker Hub, ECR, and GCR         | ■ Reading  | 30 min · 75 XP  |
| 23 | CI/CD Pipelines with Docker: GitHub Actions | ■ Reading  | 35 min · 75 XP  |
| 24 | BuildKit and Advanced Build Features        | ■ Reading  | 35 min · 75 XP  |
| 25 | Registry and CI/CD Practice                 | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 26 | Docker Swarm: Orchestration Basics                     | ■ Reading  | 35 min · 75 XP  |
| 27 | Transitioning from Docker Swarm to Kubernetes          | ■ Reading  | 35 min · 75 XP  |
| 28 | Debugging Containers: exec, logs, and inspect          | ■ Reading  | 30 min · 75 XP  |
| 29 | Performance: Image Size Optimisation and Layer Caching | ■ Reading  | 35 min · 75 XP  |
| 30 | Orchestration and Debugging Practice                   | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                       |           |                |
|----|---------------------------------------|-----------|----------------|
| 31 | Podman and OCI Alternatives to Docker | ■ Reading | 30 min · 75 XP |
|----|---------------------------------------|-----------|----------------|

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 32 | Dev Containers: VS Code and Codespaces            | ■ Reading | 30 min · 75 XP  |
| 33 | Monitoring Containers: cAdvisor and Prometheus    | ■ Reading | 35 min · 75 XP  |
| 34 | Docker in Production: Rollouts and Rollbacks      | ■ Reading | 35 min · 75 XP  |
| 35 | Capstone: Container-Native Microservices Platform | ■ Project | 60 min · 150 XP |

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# Kubernetes in Production

■ Pending

Slug: **kubernetes** · Difficulty: **Advanced** · Duration: **48h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Pods, Helm, Operators, RBAC, GitOps, Istio

## Module 1

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Kubernetes: Architecture and Components | ■ Reading  | 30 min · 75 XP  |
| 02 | Pods: The Smallest Deployable Unit                      | ■ Reading  | 35 min · 75 XP  |
| 03 | Deployments: Replicas, Rolling Updates, and Rollbacks   | ■ Reading  | 35 min · 75 XP  |
| 04 | Services: ClusterIP, NodePort, and LoadBalancer         | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Deploy a Multi-Tier Application    | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 06 | ConfigMaps and Secrets: Configuration Management | ■ Reading  | 35 min · 75 XP  |
| 07 | Namespaces and Resource Quotas                   | ■ Reading  | 35 min · 75 XP  |
| 08 | Ingress Controllers and TLS Termination          | ■ Reading  | 35 min · 75 XP  |
| 09 | Persistent Volumes and Storage Classes           | ■ Reading  | 35 min · 75 XP  |
| 10 | Config and Storage Practice                      | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 11 | StatefulSets for Stateful Applications                | ■ Reading  | 35 min · 75 XP  |
| 12 | DaemonSets and Jobs: Cluster-Wide and Batch Workloads | ■ Reading  | 35 min · 75 XP  |
| 13 | Horizontal Pod Autoscaler and Cluster Autoscaler      | ■ Reading  | 35 min · 75 XP  |
| 14 | Resource Requests and Limits: QoS Classes             | ■ Reading  | 35 min · 75 XP  |
| 15 | Workloads Assessment: Stateful Database Cluster       | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                       |           |                 |
|----|-------------------------------------------------------|-----------|-----------------|
| 16 | RBAC: Roles, ClusterRoles, and Bindings               | ■ Reading | 35 min · 75 XP  |
| 17 | ServiceAccounts and Pod Security Standards            | ■ Reading | 35 min · 75 XP  |
| 18 | NetworkPolicies: Segmenting Pod Traffic               | ■ Reading | 35 min · 75 XP  |
| 19 | Secrets Management with External Secrets Operator     | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Production Multi-Tier Application | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | Helm: Packaging and Deploying Kubernetes Applications | ■ Reading  | 35 min · 75 XP  |
| 22 | Helm Charts: Writing, Testing, and Publishing         | ■ Reading  | 35 min · 75 XP  |
| 23 | Operators: Custom Resources and Controllers           | ■ Reading  | 35 min · 75 XP  |
| 24 | GitOps with Argo CD: Declarative Deployments          | ■ Reading  | 35 min · 75 XP  |
| 25 | Helm and GitOps Practice                              | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 26 | Istio Service Mesh: Traffic Management and mTLS | ■ Reading  | 35 min · 75 XP  |
| 27 | Observability: Prometheus, Grafana, and Loki    | ■ Reading  | 35 min · 75 XP  |
| 28 | Distributed Tracing with Jaeger in Kubernetes   | ■ Reading  | 35 min · 75 XP  |
| 29 | Cluster Upgrades and Node Maintenance           | ■ Reading  | 35 min · 75 XP  |
| 30 | Service Mesh and Observability Practice         | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                             |           |                |
|----|-------------------------------------------------------------|-----------|----------------|
| 31 | Multi-Cluster Kubernetes: Federation and Argo CD            | ■ Reading | 35 min · 75 XP |
| 32 | Cost Optimisation: Spot Instances and Resource Right-Sizing | ■ Reading | 35 min · 75 XP |
| 33 | Disaster Recovery and Backup with Velero                    | ■ Reading | 35 min · 75 XP |
| 34 | CKA Exam Preparation and Tips                               | ■ Reading | 35 min · 75 XP |





# Terraform & IaC

■ Pending

Slug: **terraform** · Difficulty: **Intermediate** · Duration: **28h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: HCL, Modules, Remote State, Multi-Cloud

## Module 1

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 01 | Introduction to IaC and Terraform Architecture | ■ Reading  | 30 min · 75 XP  |
| 02 | HCL Syntax: Resources, Variables, and Outputs  | ■ Reading  | 30 min · 75 XP  |
| 03 | Providers: AWS, GCP, Azure, and Kubernetes     | ■ Reading  | 35 min · 75 XP  |
| 04 | Terraform Workflow: init, plan, apply, destroy | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: AWS VPC with Terraform    | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 06 | Variables: Types, Validation, and Sensitive Values    | ■ Reading  | 35 min · 75 XP  |
| 07 | Locals, Data Sources, and Dynamic Blocks              | ■ Reading  | 35 min · 75 XP  |
| 08 | State: Local, Remote, and State Locking               | ■ Reading  | 35 min · 75 XP  |
| 09 | Workspaces: Environment Separation                    | ■ Reading  | 30 min · 75 XP  |
| 10 | State and Variables Practice: Multi-Environment Setup | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 11 | Modules: Writing, Using, and Publishing               | ■ Reading  | 35 min · 75 XP  |
| 12 | Module Composition and Dependency Management          | ■ Reading  | 35 min · 75 XP  |
| 13 | Terraform Registry and Community Modules              | ■ Reading  | 30 min · 75 XP  |
| 14 | Count, For Each, and Dynamic Resource Creation        | ■ Reading  | 35 min · 75 XP  |
| 15 | Modules Practice: Reusable AWS Infrastructure Library | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 16 | Provisioners: Local-Exec and Remote-Exec          | ■ Reading | 35 min · 75 XP  |
| 17 | Null Resources and Triggers                       | ■ Reading | 30 min · 75 XP  |
| 18 | Managing Kubernetes with the Kubernetes Provider  | ■ Reading | 35 min · 75 XP  |
| 19 | Helm Provider: Deploying Charts with Terraform    | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Full AWS Infrastructure Stack | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 21 | Remote State with S3, DynamoDB, and Terraform Cloud | ■ Reading  | 35 min · 75 XP  |
| 22 | Terragrunt: DRY Terraform at Scale                  | ■ Reading  | 35 min · 75 XP  |
| 23 | Atlantis: Automated Terraform Pull Requests         | ■ Reading  | 35 min · 75 XP  |
| 24 | CI/CD for Terraform: GitHub Actions and Pipelines   | ■ Reading  | 35 min · 75 XP  |
| 25 | Remote State and CI/CD Practice                     | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 26 | Policy as Code: Sentinel and Open Policy Agent       | ■ Reading  | 35 min · 75 XP  |
| 27 | Drift Detection and Reconciliation                   | ■ Reading  | 35 min · 75 XP  |
| 28 | Testing Terraform: Terratest and tftest              | ■ Reading  | 35 min · 75 XP  |
| 29 | Security: Secrets in Terraform and Vault Integration | ■ Reading  | 35 min · 75 XP  |
| 30 | Policy, Testing, and Security Practice               | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                       |           |                |
|----|-------------------------------------------------------|-----------|----------------|
| 31 | Refactoring Terraform: Moved Blocks and State Surgery | ■ Reading | 35 min · 75 XP |
| 32 | Multi-Cloud Deployments with Terraform                | ■ Reading | 35 min · 75 XP |
| 33 | Cost Estimation with Infracost                        | ■ Reading | 30 min · 75 XP |
| 34 | Terraform CDK: IaC with TypeScript and Python         | ■ Reading | 35 min · 75 XP |



# AI & Machine Learning

## Python for AI & ML

■ Complete

Slug: `python-for-ai-ml` · Difficulty: **Beginner** · Duration: 45h planned · 22h lessons · Lessons: 35 (■29 ■4 ■0 ■2) · Tags: Python, NumPy, Pandas, Scikit-Learn, Deep Learning

### Module 1

|    |                                                      |           |                |
|----|------------------------------------------------------|-----------|----------------|
| 01 | Introduction to Python and AI/ML Fundamentals        | ■ Reading | 30 min · 75 XP |
| 02 | Python Basics: Variables, Data Types, and Operations | ■ Reading | 30 min · 75 XP |
| 03 | Control Flow: Conditionals and Loops                 | ■ Reading | 30 min · 75 XP |
| 04 | Functions and Modules in Python                      | ■ Reading | 30 min · 75 XP |
| 05 | Working with Lists, Tuples, and Dictionaries         | ■ Reading | 30 min · 75 XP |

### Module 2

|    |                                           |            |                 |
|----|-------------------------------------------|------------|-----------------|
| 06 | File Handling and Input/Output Operations | ■ Reading  | 30 min · 75 XP  |
| 07 | Object-Oriented Programming Fundamentals  | ■ Reading  | 35 min · 75 XP  |
| 08 | Exception Handling and Debugging          | ■ Reading  | 30 min · 75 XP  |
| 09 | NumPy: Creating and Manipulating Arrays   | ■ Reading  | 35 min · 75 XP  |
| 10 | Foundation Concepts Review and Practice   | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                             |           |                |
|----|---------------------------------------------|-----------|----------------|
| 11 | Pandas: Data Structures and Loading Data    | ■ Reading | 35 min · 75 XP |
| 12 | Exploratory Data Analysis with Pandas       | ■ Reading | 35 min · 75 XP |
| 13 | Data Cleaning and Preprocessing Techniques  | ■ Reading | 35 min · 75 XP |
| 14 | Feature Engineering and Selection           | ■ Reading | 35 min · 75 XP |
| 15 | Statistical Analysis and Hypothesis Testing | ■ Reading | 35 min · 75 XP |

### Module 4

|    |                                                             |            |                 |
|----|-------------------------------------------------------------|------------|-----------------|
| 16 | Data Visualization with Matplotlib and Seaborn              | ■ Reading  | 35 min · 75 XP  |
| 17 | Introduction to Scikit-Learn and Model Building             | ■ Reading  | 35 min · 75 XP  |
| 18 | Core Skills Assessment and Practice                         | ■ Exercise | 50 min · 100 XP |
| 19 | Supervised Learning: Regression Models                      | ■ Reading  | 35 min · 75 XP  |
| 20 | End-to-End Machine Learning Project: House Price Prediction | ■ Project  | 60 min · 150 XP |

### Module 5

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 21 | Supervised Learning: Classification Models    | ■ Reading  | 35 min · 75 XP  |
| 22 | Model Evaluation Metrics and Cross-Validation | ■ Reading  | 35 min · 75 XP  |
| 23 | Hyperparameter Tuning and Grid Search         | ■ Reading  | 35 min · 75 XP  |
| 24 | Ensemble Methods: Bagging and Boosting        | ■ Reading  | 35 min · 75 XP  |
| 25 | Intermediate Machine Learning Practice        | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 26 | Unsupervised Learning: Clustering Algorithms       | ■ Reading  | 35 min · 75 XP  |
| 27 | Dimensionality Reduction Techniques                | ■ Reading  | 35 min · 75 XP  |
| 28 | Introduction to Neural Networks and Deep Learning  | ■ Reading  | 35 min · 75 XP  |
| 29 | Building Neural Networks with TensorFlow and Keras | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Topics Assessment and Practice            | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                                    |           |                |
|----|----------------------------------------------------|-----------|----------------|
| 31 | Convolutional Neural Networks for Image Processing | ■ Reading | 35 min · 75 XP |
|----|----------------------------------------------------|-----------|----------------|

|    |                                                 |           |                 |
|----|-------------------------------------------------|-----------|-----------------|
| 32 | Recurrent Neural Networks and Sequence Modeling | ■ Reading | 35 min · 75 XP  |
| 33 | Natural Language Processing with Python         | ■ Reading | 35 min · 75 XP  |
| 34 | Model Deployment and Production Considerations  | ■ Reading | 35 min · 75 XP  |
| 35 | Comprehensive AI/ML Capstone Project            | ■ Project | 60 min · 150 XP |

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# PyTorch Deep Learning

■ Complete

Slug: [pytorch-deep-learning](#) · Difficulty: **Intermediate** · Duration: **55h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: PyTorch, CNNs, RNNs, GANs, Deployment

## Module 1

|    |                                                         |           |                |
|----|---------------------------------------------------------|-----------|----------------|
| 01 | Introduction to PyTorch and the Deep Learning Ecosystem | ■ Reading | 30 min · 75 XP |
| 02 | Tensors: Creation, Operations, and GPU Acceleration     | ■ Reading | 35 min · 75 XP |
| 03 | Autograd and Automatic Differentiation                  | ■ Reading | 35 min · 75 XP |
| 04 | Building Neural Networks with nn.Module                 | ■ Reading | 35 min · 75 XP |
| 05 | Loss Functions and Optimizers                           | ■ Reading | 30 min · 75 XP |

## Module 2

|    |                                          |            |                 |
|----|------------------------------------------|------------|-----------------|
| 06 | Training and Validation Loops            | ■ Reading  | 35 min · 75 XP  |
| 07 | Data Loading with Dataset and DataLoader | ■ Reading  | 35 min · 75 XP  |
| 08 | Transfer Learning with Pretrained Models | ■ Reading  | 35 min · 75 XP  |
| 09 | Convolutional Neural Networks (CNNs)     | ■ Reading  | 35 min · 75 XP  |
| 10 | Foundational Skills Exercise             | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                   |           |                |
|----|---------------------------------------------------|-----------|----------------|
| 11 | Recurrent Neural Networks and LSTMs               | ■ Reading | 35 min · 75 XP |
| 12 | Attention Mechanisms and Transformers in PyTorch  | ■ Reading | 35 min · 75 XP |
| 13 | Custom Loss Functions and Metrics                 | ■ Reading | 35 min · 75 XP |
| 14 | Regularization: Dropout, Batch Norm, Weight Decay | ■ Reading | 35 min · 75 XP |
| 15 | Learning Rate Scheduling and Warm-Up Strategies   | ■ Reading | 35 min · 75 XP |

## Module 4

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 16 | Mixed Precision Training with torch.cuda.amp      | ■ Reading  | 35 min · 75 XP  |
| 17 | Distributed Training with DistributedDataParallel | ■ Reading  | 35 min · 75 XP  |
| 18 | Intermediate Skills Checkpoint                    | ■ Exercise | 50 min · 100 XP |
| 19 | PyTorch Lightning for Clean Training Code         | ■ Reading  | 35 min · 75 XP  |
| 20 | End-to-End Project: Image Classification Pipeline | ■ Project  | 60 min · 150 XP |

## Module 5

|    |                                        |            |                 |
|----|----------------------------------------|------------|-----------------|
| 21 | Object Detection with PyTorch          | ■ Reading  | 35 min · 75 XP  |
| 22 | Semantic Segmentation Models           | ■ Reading  | 35 min · 75 XP  |
| 23 | Time Series Forecasting with PyTorch   | ■ Reading  | 35 min · 75 XP  |
| 24 | Generative Adversarial Networks (GANs) | ■ Reading  | 35 min · 75 XP  |
| 25 | Advanced Techniques Exercise           | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 26 | Variational Autoencoders (VAEs)               | ■ Reading  | 35 min · 75 XP  |
| 27 | Model Pruning and Quantization                | ■ Reading  | 35 min · 75 XP  |
| 28 | TorchScript and Model Export                  | ■ Reading  | 35 min · 75 XP  |
| 29 | ONNX Export and Cross-Platform Deployment     | ■ Reading  | 35 min · 75 XP  |
| 30 | Optimization Exercise: Latency and Throughput | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                        |           |                |
|----|----------------------------------------|-----------|----------------|
| 31 | PyTorch Mobile and Edge Deployment     | ■ Reading | 35 min · 75 XP |
| 32 | Custom CUDA Extensions                 | ■ Reading | 35 min · 75 XP |
| 33 | Debugging and Profiling PyTorch Models | ■ Reading | 35 min · 75 XP |
| 34 | Testing Neural Networks Effectively    | ■ Reading | 35 min · 75 XP |



# TensorFlow & Keras

■ Complete

Slug: tensorflow-keras · Difficulty: **Intermediate** · Duration: 50h planned · 21h lessons · Lessons: 35 (■29 ■4 ■0 ■2) ·

Tags: TensorFlow, Keras, CNNs, Transfer Learning, TFLite

## Module 1

|    |                                                      |           |                |
|----|------------------------------------------------------|-----------|----------------|
| 01 | Introduction to TensorFlow and Keras Ecosystem       | ■ Reading | 30 min · 75 XP |
| 02 | Understanding Tensors and Data Structures            | ■ Reading | 30 min · 75 XP |
| 03 | Setting Up Your TensorFlow Development Environment   | ■ Reading | 20 min · 50 XP |
| 04 | NumPy Fundamentals for TensorFlow                    | ■ Reading | 30 min · 75 XP |
| 05 | Computational Graphs and Static vs Dynamic Execution | ■ Reading | 35 min · 75 XP |

## Module 2

|    |                                            |            |                 |
|----|--------------------------------------------|------------|-----------------|
| 06 | Keras Sequential API Basics                | ■ Reading  | 30 min · 75 XP  |
| 07 | Dense Layers and Activation Functions      | ■ Reading  | 30 min · 75 XP  |
| 08 | Loss Functions and Optimization Algorithms | ■ Reading  | 35 min · 75 XP  |
| 09 | Training, Validation, and Test Sets        | ■ Reading  | 30 min · 75 XP  |
| 10 | Foundation Concepts Review and Practice    | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                              |           |                |
|----|----------------------------------------------|-----------|----------------|
| 11 | Convolutional Neural Networks Architecture   | ■ Reading | 35 min · 75 XP |
| 12 | Pooling and Flattening Operations            | ■ Reading | 30 min · 75 XP |
| 13 | Regularization Techniques: Dropout and L1/L2 | ■ Reading | 35 min · 75 XP |
| 14 | Batch Normalization and Layer Normalization  | ■ Reading | 35 min · 75 XP |
| 15 | Recurrent Neural Networks and LSTMs          | ■ Reading | 35 min · 75 XP |

## Module 4

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 16 | GRU Cells and Bidirectional RNNs               | ■ Reading  | 35 min · 75 XP  |
| 17 | Transfer Learning with Pretrained Models       | ■ Reading  | 35 min · 75 XP  |
| 18 | Core Skills Application Exercise               | ■ Exercise | 50 min · 100 XP |
| 19 | Building Custom Layers and Models              | ■ Reading  | 35 min · 75 XP  |
| 20 | Image Classification Project: CIFAR-10 Dataset | ■ Project  | 60 min · 150 XP |

## Module 5

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 21 | Data Augmentation and Preprocessing Pipelines   | ■ Reading  | 35 min · 75 XP  |
| 22 | Hyperparameter Tuning and Grid Search           | ■ Reading  | 35 min · 75 XP  |
| 23 | Early Stopping and Learning Rate Scheduling     | ■ Reading  | 35 min · 75 XP  |
| 24 | Model Evaluation Metrics and Confusion Matrices | ■ Reading  | 35 min · 75 XP  |
| 25 | Core Skills Mastery Exercise                    | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 26 | Functional API for Complex Model Architectures | ■ Reading  | 35 min · 75 XP  |
| 27 | Multi-Input and Multi-Output Models            | ■ Reading  | 35 min · 75 XP  |
| 28 | Attention Mechanisms and Transformers          | ■ Reading  | 35 min · 75 XP  |
| 29 | Natural Language Processing with Keras         | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Techniques Application Exercise       | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                           |           |                |
|----|-------------------------------------------|-----------|----------------|
| 31 | Model Deployment and TensorFlow Lite      | ■ Reading | 35 min · 75 XP |
| 32 | Distributed Training and Data Parallelism | ■ Reading | 35 min · 75 XP |
| 33 | Custom Training Loops with GradientTape   | ■ Reading | 35 min · 75 XP |
| 34 | Debugging and Profiling TensorFlow Models | ■ Reading | 35 min · 75 XP |





# Hugging Face Transformers

■ Complete

Slug: [hugging-face-transformers](#) · Difficulty: **Intermediate** · Duration: **45h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: BERT, GPT, Fine-tuning, Tokenizers, HF Hub

## Module 1

|    |                                           |           |                |
|----|-------------------------------------------|-----------|----------------|
| 01 | Introduction to Transformers Architecture | ■ Reading | 30 min · 75 XP |
| 02 | Attention Mechanism Fundamentals          | ■ Reading | 35 min · 75 XP |
| 03 | Self-Attention and Multi-Head Attention   | ■ Reading | 35 min · 75 XP |
| 04 | Encoder-Decoder Architecture Overview     | ■ Reading | 35 min · 75 XP |
| 05 | Understanding BERT and Its Variants       | ■ Reading | 35 min · 75 XP |

## Module 2

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 06 | GPT Models and Autoregressive Language Modeling  | ■ Reading  | 35 min · 75 XP  |
| 07 | Tokenization Strategies and Subword Tokenization | ■ Reading  | 35 min · 75 XP  |
| 08 | Positional Encoding and Embeddings               | ■ Reading  | 35 min · 75 XP  |
| 09 | Transfer Learning with Pretrained Models         | ■ Reading  | 35 min · 75 XP  |
| 10 | Foundation Concepts Review and Practice          | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                      |           |                |
|----|------------------------------------------------------|-----------|----------------|
| 11 | Setting Up Hugging Face Environment and Installation | ■ Reading | 20 min · 50 XP |
| 12 | Loading and Using Pretrained Models                  | ■ Reading | 30 min · 75 XP |
| 13 | Working with Tokenizers from the Hub                 | ■ Reading | 30 min · 75 XP |
| 14 | Fine-tuning BERT for Text Classification             | ■ Reading | 35 min · 75 XP |
| 15 | Named Entity Recognition with Transformers           | ■ Reading | 35 min · 75 XP |

## Module 4

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 16 | Question Answering with SQuAD Dataset       | ■ Reading  | 35 min · 75 XP  |
| 17 | Semantic Similarity and Sentence Embeddings | ■ Reading  | 35 min · 75 XP  |
| 18 | Core Skills Checkpoint Exercise             | ■ Exercise | 50 min · 100 XP |
| 19 | Sequence-to-Sequence Models for Translation | ■ Reading  | 35 min · 75 XP  |
| 20 | Building a Complete NLP Pipeline Project    | ■ Project  | 60 min · 150 XP |

## Module 5

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 21 | Custom Training Loops with PyTorch               | ■ Reading  | 35 min · 75 XP  |
| 22 | Distributed Training and Mixed Precision         | ■ Reading  | 35 min · 75 XP  |
| 23 | Optimizing Model Performance and Inference Speed | ■ Reading  | 35 min · 75 XP  |
| 24 | Model Quantization and Pruning Techniques        | ■ Reading  | 35 min · 75 XP  |
| 25 | Advanced Skills Assessment                       | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 26 | Zero-Shot and Few-Shot Learning Strategies       | ■ Reading  | 35 min · 75 XP  |
| 27 | Prompt Engineering and In-Context Learning       | ■ Reading  | 35 min · 75 XP  |
| 28 | Multi-Task Learning with Transformers            | ■ Reading  | 35 min · 75 XP  |
| 29 | Domain Adaptation and Transfer Learning Advanced | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Optimization Techniques                 | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                  |           |                |
|----|--------------------------------------------------|-----------|----------------|
| 31 | Vision Transformers and Multimodal Models        | ■ Reading | 35 min · 75 XP |
| 32 | Efficient Transformers and Long-Context Modeling | ■ Reading | 35 min · 75 XP |
| 33 | Deploying Models to Production with HF Hub       | ■ Reading | 35 min · 75 XP |
| 34 | Monitoring and Evaluating Model Performance      | ■ Reading | 35 min · 75 XP |



# Large Language Models

■ Complete

Slug: [large-language-models](#) · Difficulty: **Advanced** · Duration: **60h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: LLMs, Transformers, RLHF, RAG, Deployment

## Module 1

|    |                                                 |           |                |
|----|-------------------------------------------------|-----------|----------------|
| 01 | Introduction to Large Language Models           | ■ Reading | 30 min · 75 XP |
| 02 | History and Evolution of Neural Language Models | ■ Reading | 30 min · 75 XP |
| 03 | Fundamentals of Natural Language Processing     | ■ Reading | 30 min · 75 XP |
| 04 | Understanding Transformer Architecture          | ■ Reading | 35 min · 75 XP |
| 05 | Attention Mechanisms Explained                  | ■ Reading | 35 min · 75 XP |

## Module 2

|    |                                            |            |                 |
|----|--------------------------------------------|------------|-----------------|
| 06 | Word Embeddings and Vector Representations | ■ Reading  | 35 min · 75 XP  |
| 07 | Tokenization and Preprocessing Techniques  | ■ Reading  | 30 min · 75 XP  |
| 08 | Training Data and Dataset Considerations   | ■ Reading  | 35 min · 75 XP  |
| 09 | Loss Functions and Optimization in LLMs    | ■ Reading  | 35 min · 75 XP  |
| 10 | Foundation Concepts Assessment             | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                |           |                |
|----|------------------------------------------------|-----------|----------------|
| 11 | Transfer Learning and Pre-training Strategies  | ■ Reading | 35 min · 75 XP |
| 12 | Fine-tuning Language Models for Specific Tasks | ■ Reading | 35 min · 75 XP |
| 13 | Prompt Engineering Fundamentals                | ■ Reading | 30 min · 75 XP |
| 14 | Few-shot and Zero-shot Learning                | ■ Reading | 35 min · 75 XP |
| 15 | Chain of Thought Prompting Techniques          | ■ Reading | 35 min · 75 XP |

## Module 4

|    |                                           |            |                 |
|----|-------------------------------------------|------------|-----------------|
| 16 | Model Evaluation Metrics and Benchmarks   | ■ Reading  | 35 min · 75 XP  |
| 17 | Scaling Laws and Model Size Trade-offs    | ■ Reading  | 35 min · 75 XP  |
| 18 | Core Skills Practice Exercise             | ■ Exercise | 50 min · 100 XP |
| 19 | Inference Optimization and Quantization   | ■ Reading  | 35 min · 75 XP  |
| 20 | Building LLM-powered Applications Project | ■ Project  | 60 min · 150 XP |

## Module 5

|    |                                          |            |                 |
|----|------------------------------------------|------------|-----------------|
| 21 | API Integration and Deployment           | ■ Reading  | 35 min · 75 XP  |
| 22 | Safety, Bias, and Ethical Considerations | ■ Reading  | 35 min · 75 XP  |
| 23 | Hallucinations and Reliability Issues    | ■ Reading  | 35 min · 75 XP  |
| 24 | Retrieval-Augmented Generation (RAG)     | ■ Reading  | 35 min · 75 XP  |
| 25 | Advanced Skills Consolidation Exercise   | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                          |            |                 |
|----|----------------------------------------------------------|------------|-----------------|
| 26 | Multi-modal Language Models                              | ■ Reading  | 35 min · 75 XP  |
| 27 | Parameter-Efficient Fine-tuning Methods                  | ■ Reading  | 35 min · 75 XP  |
| 28 | Mixture of Experts and Sparse Models                     | ■ Reading  | 35 min · 75 XP  |
| 29 | Alignment and Reinforcement Learning from Human Feedback | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Topics Application Exercise                     | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                              |           |                |
|----|----------------------------------------------|-----------|----------------|
| 31 | Emerging Architectures and Future Directions | ■ Reading | 35 min · 75 XP |
| 32 | Efficient Inference at Scale                 | ■ Reading | 35 min · 75 XP |
| 33 | LLM Interpretability and Analysis            | ■ Reading | 35 min · 75 XP |
| 34 | Building Production-Ready LLM Systems        | ■ Reading | 35 min · 75 XP |



# Retrieval-Augmented Generation

■ Complete

Slug: **retrieval-augmented-generation** · Difficulty: **Advanced** · Duration: **40h planned** · **22h lessons** · Lessons: **35** (■29 ■4 ■0 ■2) · Tags: RAG, Embeddings, Vector DBs, LangChain, RAGAS

## Module 1

|    |                                                  |           |                |
|----|--------------------------------------------------|-----------|----------------|
| 01 | Introduction to RAG and Knowledge Retrieval      | ■ Reading | 30 min · 75 XP |
| 02 | Embeddings and Vector Representations            | ■ Reading | 35 min · 75 XP |
| 03 | Vector Databases: Pinecone, Weaviate, and Chroma | ■ Reading | 35 min · 75 XP |
| 04 | Document Chunking Strategies                     | ■ Reading | 35 min · 75 XP |
| 05 | Semantic Search and Similarity Metrics           | ■ Reading | 35 min · 75 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | Building a Basic RAG Pipeline                     | ■ Reading  | 35 min · 75 XP  |
| 07 | Retrieval Evaluation: Precision, Recall, and NDCG | ■ Reading  | 35 min · 75 XP  |
| 08 | Hybrid Search: BM25 and Dense Retrieval           | ■ Reading  | 35 min · 75 XP  |
| 09 | Multi-Stage Retrieval and Reranking               | ■ Reading  | 35 min · 75 XP  |
| 10 | Core RAG Skills Exercise                          | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                              |           |                |
|----|----------------------------------------------|-----------|----------------|
| 11 | Advanced Chunking: Semantic and Hierarchical | ■ Reading | 35 min · 75 XP |
| 12 | Query Expansion and Reformulation Techniques | ■ Reading | 35 min · 75 XP |
| 13 | Hypothetical Document Embeddings (HyDE)      | ■ Reading | 35 min · 75 XP |
| 14 | Contextual Compression and Filtering         | ■ Reading | 35 min · 75 XP |
| 15 | Multi-Hop and Iterative Retrieval            | ■ Reading | 35 min · 75 XP |

## Module 4

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 16 | RAG Evaluation with the RAGAS Framework          | ■ Reading  | 35 min · 75 XP  |
| 17 | Handling Conflicting and Contradictory Documents | ■ Reading  | 35 min · 75 XP  |
| 18 | Intermediate RAG Checkpoint                      | ■ Exercise | 50 min · 100 XP |
| 19 | GraphRAG: Knowledge Graph Integration            | ■ Reading  | 35 min · 75 XP  |
| 20 | End-to-End Project: Document Q&A; System         | ■ Project  | 60 min · 150 XP |

## Module 5

|    |                                     |            |                 |
|----|-------------------------------------|------------|-----------------|
| 21 | Streaming RAG Responses             | ■ Reading  | 35 min · 75 XP  |
| 22 | Long-Context RAG with Late Chunking | ■ Reading  | 35 min · 75 XP  |
| 23 | Domain-Specific RAG Fine-Tuning     | ■ Reading  | 35 min · 75 XP  |
| 24 | Cache-Augmented Generation          | ■ Reading  | 35 min · 75 XP  |
| 25 | Advanced RAG Patterns Exercise      | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 26 | Production RAG Architecture and Scalability | ■ Reading  | 35 min · 75 XP  |
| 27 | Indexing Large Corpora Efficiently          | ■ Reading  | 35 min · 75 XP  |
| 28 | Self-Corrective RAG (CRAG)                  | ■ Reading  | 35 min · 75 XP  |
| 29 | Agentic RAG: Dynamic and Adaptive Retrieval | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Production Exercise                | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                          |           |                |
|----|------------------------------------------|-----------|----------------|
| 31 | Multimodal RAG: Images, Tables, and PDFs | ■ Reading | 35 min · 75 XP |
| 32 | Real-Time RAG with Streaming Ingestion   | ■ Reading | 35 min · 75 XP |
| 33 | RAG Security and PII Handling            | ■ Reading | 35 min · 75 XP |
| 34 | Cost Optimization for RAG Pipelines      | ■ Reading | 35 min · 75 XP |



# AI Agents & Agentic Workflows

■ Complete

Slug: ai-agents-agentic-workflows · Difficulty: **Advanced** · Duration: 50h planned · 22h lessons · Lessons: 35 (■29 ■4 ■0 ■2) · Tags: Agents, ReAct, LangGraph, AutoGen, CrewAI

## Module 1

|    |                                                         |           |                |
|----|---------------------------------------------------------|-----------|----------------|
| 01 | Introduction to AI Agents and Agentic Systems           | ■ Reading | 30 min · 75 XP |
| 02 | Agent Architectures: Reactive, Deliberative, and Hybrid | ■ Reading | 35 min · 75 XP |
| 03 | Memory and State Management in Agents                   | ■ Reading | 35 min · 75 XP |
| 04 | Tool Use and Function Calling                           | ■ Reading | 35 min · 75 XP |
| 05 | Planning and Goal Decomposition                         | ■ Reading | 35 min · 75 XP |

## Module 2

|    |                                              |            |                 |
|----|----------------------------------------------|------------|-----------------|
| 06 | The ReAct Pattern: Reasoning and Acting      | ■ Reading  | 35 min · 75 XP  |
| 07 | Chain of Thought for Agent Reasoning         | ■ Reading  | 35 min · 75 XP  |
| 08 | Multi-Agent Systems Overview                 | ■ Reading  | 35 min · 75 XP  |
| 09 | Agent Communication Protocols                | ■ Reading  | 35 min · 75 XP  |
| 10 | Core Concepts Exercise: Build a Simple Agent | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                        |           |                |
|----|----------------------------------------|-----------|----------------|
| 11 | LangChain Agents Deep Dive             | ■ Reading | 35 min · 75 XP |
| 12 | LangGraph for Stateful Agents          | ■ Reading | 35 min · 75 XP |
| 13 | AutoGen: Multi-Agent Conversations     | ■ Reading | 35 min · 75 XP |
| 14 | CrewAI: Role-Based Agent Collaboration | ■ Reading | 35 min · 75 XP |
| 15 | Custom Tool Development for Agents     | ■ Reading | 35 min · 75 XP |

## Module 4

|    |                                             |            |                 |
|----|---------------------------------------------|------------|-----------------|
| 16 | Context Window Management and Summarization | ■ Reading  | 35 min · 75 XP  |
| 17 | Agent Orchestration and Delegation Patterns | ■ Reading  | 35 min · 75 XP  |
| 18 | Intermediate Skills Checkpoint              | ■ Exercise | 50 min · 100 XP |
| 19 | Advanced Planning: Tree of Thoughts         | ■ Reading  | 35 min · 75 XP  |
| 20 | End-to-End Project: Research Agent          | ■ Project  | 60 min · 150 XP |

## Module 5

|    |                                              |            |                 |
|----|----------------------------------------------|------------|-----------------|
| 21 | Agent Evaluation and Benchmarking            | ■ Reading  | 35 min · 75 XP  |
| 22 | Error Handling and Agent Recovery Strategies | ■ Reading  | 35 min · 75 XP  |
| 23 | Human-in-the-Loop Agent Design               | ■ Reading  | 35 min · 75 XP  |
| 24 | Agent Security and Trust Boundaries          | ■ Reading  | 35 min · 75 XP  |
| 25 | Advanced Agent Patterns Exercise             | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                         |            |                 |
|----|-----------------------------------------|------------|-----------------|
| 26 | Production Deployment of AI Agents      | ■ Reading  | 35 min · 75 XP  |
| 27 | Monitoring and Observability for Agents | ■ Reading  | 35 min · 75 XP  |
| 28 | Cost Optimization for Agent Pipelines   | ■ Reading  | 35 min · 75 XP  |
| 29 | Agent Testing and Quality Assurance     | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Deployment Exercise            | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                  |           |                |
|----|--------------------------------------------------|-----------|----------------|
| 31 | Multimodal Agents: Vision and Audio Capabilities | ■ Reading | 35 min · 75 XP |
| 32 | Autonomous Code Generation Agents                | ■ Reading | 35 min · 75 XP |
| 33 | Domain-Specific Agent Customization              | ■ Reading | 35 min · 75 XP |
| 34 | Enterprise Agent Architecture Patterns           | ■ Reading | 35 min · 75 XP |





# Machine Learning Fundamentals

■ Pending

Slug: [ml-fundamentals](#) · Difficulty: **Intermediate** · Duration: **38h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Regression, Classification, Clustering, Scikit-Learn

## Module 1

|    |                                                          |            |                 |
|----|----------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Machine Learning: Types and Applications | ■ Reading  | 30 min · 75 XP  |
| 02 | Python for ML: NumPy, Pandas, and Matplotlib Recap       | ■ Reading  | 30 min · 75 XP  |
| 03 | Data Preprocessing: Cleaning, Encoding, and Scaling      | ■ Reading  | 35 min · 75 XP  |
| 04 | Feature Engineering: Selection and Transformation        | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: EDA and Preprocessing Pipeline      | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 06 | Linear Regression: Theory and Implementation      | ■ Reading  | 35 min · 75 XP  |
| 07 | Polynomial Regression and Regularisation          | ■ Reading  | 35 min · 75 XP  |
| 08 | Logistic Regression for Classification            | ■ Reading  | 35 min · 75 XP  |
| 09 | Model Evaluation: Accuracy, Precision, Recall, F1 | ■ Reading  | 35 min · 75 XP  |
| 10 | Regression and Classification Practice            | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 11 | Decision Trees and Random Forests              | ■ Reading  | 35 min · 75 XP  |
| 12 | Gradient Boosting: XGBoost, LightGBM, CatBoost | ■ Reading  | 35 min · 75 XP  |
| 13 | Support Vector Machines                        | ■ Reading  | 35 min · 75 XP  |
| 14 | k-Nearest Neighbours and Naive Bayes           | ■ Reading  | 35 min · 75 XP  |
| 15 | Ensemble Methods Assessment                    | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                            |           |                 |
|----|------------------------------------------------------------|-----------|-----------------|
| 16 | Cross-Validation and Hyperparameter Tuning                 | ■ Reading | 35 min · 75 XP  |
| 17 | Overfitting, Underfitting, and the Bias-Variance Trade-off | ■ Reading | 35 min · 75 XP  |
| 18 | Pipelines with scikit-learn                                | ■ Reading | 35 min · 75 XP  |
| 19 | Unsupervised Learning: K-Means and DBSCAN                  | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: End-to-End Classification System       | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 21 | Dimensionality Reduction: PCA and t-SNE       | ■ Reading  | 35 min · 75 XP  |
| 22 | Anomaly Detection Techniques                  | ■ Reading  | 35 min · 75 XP  |
| 23 | Time Series Analysis and Forecasting          | ■ Reading  | 35 min · 75 XP  |
| 24 | Natural Language Processing with scikit-learn | ■ Reading  | 35 min · 75 XP  |
| 25 | Unsupervised and NLP Practice                 | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 26 | Introduction to Neural Networks with scikit-learn | ■ Reading  | 35 min · 75 XP  |
| 27 | Model Explainability: SHAP and LIME               | ■ Reading  | 35 min · 75 XP  |
| 28 | ML Experiment Tracking with MLflow                | ■ Reading  | 35 min · 75 XP  |
| 29 | Model Deployment: Saving, Loading, and Serving    | ■ Reading  | 35 min · 75 XP  |
| 30 | Explainability and Deployment Practice            | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                   |           |                |
|----|---------------------------------------------------|-----------|----------------|
| 31 | ML in Production: Drift Detection and Monitoring  | ■ Reading | 35 min · 75 XP |
| 32 | Fairness, Bias, and Responsible AI                | ■ Reading | 35 min · 75 XP |
| 33 | AutoML: H2O, AutoSklearn, and TPOT                | ■ Reading | 35 min · 75 XP |
| 34 | Building an ML Platform: Tools and Infrastructure | ■ Reading | 35 min · 75 XP |



# Generative AI & LLMs

■ Pending

Slug: generative-ai · Difficulty: Intermediate · Duration: 30h planned · 22h lessons · Lessons: 35 (■28 ■5 ■0 ■2) · Tags: GPT, Fine-tuning, Embeddings, Multimodal

## Module 1

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Generative AI: GANs, VAEs, and LLMs    | ■ Reading  | 30 min · 75 XP  |
| 02 | Transformer Architecture: Attention and Self-Attention | ■ Reading  | 35 min · 75 XP  |
| 03 | Large Language Models: GPT, Llama, and Mistral         | ■ Reading  | 35 min · 75 XP  |
| 04 | Tokenization and Embeddings for Text                   | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: LLM API Integration               | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 06 | Prompt Engineering: Zero-Shot, Few-Shot, and CoT | ■ Reading  | 35 min · 75 XP  |
| 07 | Structured Output and JSON Mode                  | ■ Reading  | 35 min · 75 XP  |
| 08 | System Prompts and Conversation Management       | ■ Reading  | 35 min · 75 XP  |
| 09 | Hallucination: Causes, Detection, and Mitigation | ■ Reading  | 35 min · 75 XP  |
| 10 | Prompt Engineering Practice: AI Content Pipeline | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 11 | Fine-Tuning LLMs: When and How                  | ■ Reading  | 35 min · 75 XP  |
| 12 | LoRA and QLoRA: Parameter-Efficient Fine-Tuning | ■ Reading  | 35 min · 75 XP  |
| 13 | Instruction Tuning and RLHF                     | ■ Reading  | 35 min · 75 XP  |
| 14 | Multimodal Models: Vision-Language and Audio    | ■ Reading  | 35 min · 75 XP  |
| 15 | Fine-Tuning Assessment: Domain-Specific Model   | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                 |           |                 |
|----|-------------------------------------------------|-----------|-----------------|
| 16 | Text Embeddings: Semantic Search and Clustering | ■ Reading | 35 min · 75 XP  |
| 17 | Retrieval-Augmented Generation (RAG) Basics     | ■ Reading | 35 min · 75 XP  |
| 18 | Vector Databases: Pinecone, Weaviate, Chroma    | ■ Reading | 35 min · 75 XP  |
| 19 | Evaluation: BLEU, ROUGE, and LLM-as-Judge       | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: RAG-Powered Q&A; System     | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 21 | Building AI Applications with LangChain        | ■ Reading  | 35 min · 75 XP  |
| 22 | Image Generation: Stable Diffusion and DALL-E  | ■ Reading  | 35 min · 75 XP  |
| 23 | Code Generation Models: Copilot and Code Llama | ■ Reading  | 35 min · 75 XP  |
| 24 | AI Agents: ReAct and Tool Use                  | ■ Reading  | 35 min · 75 XP  |
| 25 | AI Application Practice: Multi-Modal Chatbot   | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 26 | Safety and Alignment: Guardrails and Red-Teaming | ■ Reading  | 35 min · 75 XP  |
| 27 | Cost Optimisation: Model Selection and Caching   | ■ Reading  | 35 min · 75 XP  |
| 28 | Deployment: LLM Serving with vLLM and Ollama     | ■ Reading  | 35 min · 75 XP  |
| 29 | Observability for AI Applications                | ■ Reading  | 35 min · 75 XP  |
| 30 | Deployment and Safety Practice                   | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                         |           |                |
|----|---------------------------------------------------------|-----------|----------------|
| 31 | AI Product Development: From Idea to Launch             | ■ Reading | 35 min · 75 XP |
| 32 | Legal and Ethical Considerations in Gen AI              | ■ Reading | 35 min · 75 XP |
| 33 | Emerging Models: Gemini, Claude, and Open-Source Trends | ■ Reading | 35 min · 75 XP |
| 34 | Multimodal Workflows: Video, Audio, and Documents       | ■ Reading | 35 min · 75 XP |



# RAG Systems & Vector Search

■ Pending

Slug: rag · Difficulty: Intermediate · Duration: 24h planned · 22h lessons · Lessons: 35 (■28 ■5 ■0 ■2) · Tags: Embeddings, Vector DB, LangChain, LlamaIndex

## Module 1

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 01 | Introduction to RAG: Architecture and Use Cases  | ■ Reading  | 30 min · 75 XP  |
| 02 | Text Embeddings: Models and Dimensionality       | ■ Reading  | 35 min · 75 XP  |
| 03 | Vector Databases: Pinecone, Chroma, and Weaviate | ■ Reading  | 35 min · 75 XP  |
| 04 | Document Loaders and Text Extraction             | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Basic Q&A; RAG Pipeline     | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 06 | Chunking Strategies: Fixed, Semantic, and Recursive | ■ Reading  | 35 min · 75 XP  |
| 07 | Embedding Models: OpenAI, Cohere, and Open-Source   | ■ Reading  | 35 min · 75 XP  |
| 08 | Similarity Search: Cosine, Dot Product, and HNSW    | ■ Reading  | 35 min · 75 XP  |
| 09 | Building a RAG Pipeline with LangChain              | ■ Reading  | 35 min · 75 XP  |
| 10 | Chunking and Retrieval Practice                     | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                       |            |                 |
|----|---------------------------------------|------------|-----------------|
| 11 | Query Preprocessing and Expansion     | ■ Reading  | 35 min · 75 XP  |
| 12 | Hybrid Search: BM25 + Dense Retrieval | ■ Reading  | 35 min · 75 XP  |
| 13 | Cross-Encoder Reranking               | ■ Reading  | 35 min · 75 XP  |
| 14 | Contextual Compression and Filtering  | ■ Reading  | 35 min · 75 XP  |
| 15 | Advanced Retrieval Practice           | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                      |           |                 |
|----|------------------------------------------------------|-----------|-----------------|
| 16 | RAG Evaluation: Faithfulness, Relevance, and Context | ■ Reading | 35 min · 75 XP  |
| 17 | The RAGAS Framework for Automated Evaluation         | ■ Reading | 35 min · 75 XP  |
| 18 | LlamaIndex: An Alternative to LangChain for RAG      | ■ Reading | 35 min · 75 XP  |
| 19 | Metadata Filtering and Structured Retrieval          | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Enterprise Document Assistant    | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                           |            |                 |
|----|-------------------------------------------|------------|-----------------|
| 21 | Multi-Hop Retrieval and Complex Queries   | ■ Reading  | 35 min · 75 XP  |
| 22 | Self-Querying Retrieval and Routing       | ■ Reading  | 35 min · 75 XP  |
| 23 | GraphRAG: Knowledge Graphs and Neo4j      | ■ Reading  | 35 min · 75 XP  |
| 24 | Agentic RAG: Dynamic Retrieval Strategies | ■ Reading  | 35 min · 75 XP  |
| 25 | Advanced RAG Patterns Practice            | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 26 | Production RAG: Scalability and Low Latency      | ■ Reading  | 35 min · 75 XP  |
| 27 | Data Ingestion Pipelines: Airflow and Prefect    | ■ Reading  | 35 min · 75 XP  |
| 28 | Multimodal RAG: Images, Tables, and PDFs         | ■ Reading  | 35 min · 75 XP  |
| 29 | Security: Access Control and PII Handling in RAG | ■ Reading  | 35 min · 75 XP  |
| 30 | Production and Security Practice                 | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                 |           |                |
|----|-------------------------------------------------|-----------|----------------|
| 31 | Cost Optimisation: Caching and Batch Processing | ■ Reading | 35 min · 75 XP |
| 32 | Streaming RAG Responses                         | ■ Reading | 35 min · 75 XP |
| 33 | Monitoring RAG Pipelines in Production          | ■ Reading | 35 min · 75 XP |
| 34 | Future of RAG: Long Context vs Retrieval        | ■ Reading | 35 min · 75 XP |



# AI Agents & Tool Use

■ Pending

Slug: **ai-agents** · Difficulty: **Advanced** · Duration: **28h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: ReAct, Function Calling, Multi-Agent, AutoGen

## Module 1

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 01 | Introduction to AI Agents: Autonomy and Decision-Making | ■ Reading  | 30 min · 75 XP  |
| 02 | Agent Architectures: Reactive, BDI, and LLM-Based       | ■ Reading  | 35 min · 75 XP  |
| 03 | The ReAct Framework: Reasoning + Acting                 | ■ Reading  | 35 min · 75 XP  |
| 04 | Tool Use and Function Calling with LLMs                 | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Build a Research Agent             | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 06 | Memory Systems: Short-Term, Long-Term, and Episodic | ■ Reading  | 35 min · 75 XP  |
| 07 | Planning and Goal Decomposition Strategies          | ■ Reading  | 35 min · 75 XP  |
| 08 | LangChain Agents: Tools, Chains, and Executors      | ■ Reading  | 35 min · 75 XP  |
| 09 | LangGraph: Stateful Agent Workflows                 | ■ Reading  | 35 min · 75 XP  |
| 10 | Memory and Planning Practice                        | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 11 | Multi-Agent Systems: Orchestration and Collaboration | ■ Reading  | 35 min · 75 XP  |
| 12 | AutoGen: Conversational Multi-Agent Frameworks       | ■ Reading  | 35 min · 75 XP  |
| 13 | CrewAI: Role-Based Agent Teams                       | ■ Reading  | 35 min · 75 XP  |
| 14 | Agent Communication Protocols and Schemas            | ■ Reading  | 35 min · 75 XP  |
| 15 | Multi-Agent Practice: Research and Writing Team      | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                  |           |                 |
|----|--------------------------------------------------|-----------|-----------------|
| 16 | Custom Tools: APIs, Code Execution, and Browsers | ■ Reading | 35 min · 75 XP  |
| 17 | Web Browsing Agents with Playwright              | ■ Reading | 35 min · 75 XP  |
| 18 | Code Generation and Execution Agents             | ■ Reading | 35 min · 75 XP  |
| 19 | Human-in-the-Loop: Approval and Feedback Loops   | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Autonomous Research Agent    | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | Agent Evaluation: Benchmarks and Success Metrics      | ■ Reading  | 35 min · 75 XP  |
| 22 | Error Handling and Agent Recovery                     | ■ Reading  | 35 min · 75 XP  |
| 23 | Context Window Management and Summarisation           | ■ Reading  | 35 min · 75 XP  |
| 24 | Agent Security: Prompt Injection and Trust Boundaries | ■ Reading  | 35 min · 75 XP  |
| 25 | Evaluation and Reliability Practice                   | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 26 | Production Deployment of AI Agents               | ■ Reading  | 35 min · 75 XP  |
| 27 | Cost Optimisation: Reducing API Calls            | ■ Reading  | 35 min · 75 XP  |
| 28 | Observability: Logging and Tracing Agent Actions | ■ Reading  | 35 min · 75 XP  |
| 29 | Testing AI Agents: Simulation and Regression     | ■ Reading  | 35 min · 75 XP  |
| 30 | Production and Testing Practice                  | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                      |           |                |
|----|------------------------------------------------------|-----------|----------------|
| 31 | Multimodal Agents: Vision and Document Understanding | ■ Reading | 35 min · 75 XP |
| 32 | Autonomous Code Review Agents                        | ■ Reading | 35 min · 75 XP |
| 33 | Domain-Specific Agents: Legal, Medical, Finance      | ■ Reading | 35 min · 75 XP |
| 34 | Future of AI Agents: AGI Pathways and Risks          | ■ Reading | 35 min · 75 XP |





# Advanced Prompt Engineering

■ Pending

Slug: `prompt-engineering` · Difficulty: **Beginner** · Duration: **12h planned** · **21h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Chain-of-Thought, Few-Shot, System Prompts

## Module 1

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Prompt Engineering: Why It Matters   | ■ Reading  | 20 min · 50 XP  |
| 02 | LLM Fundamentals: Tokens, Temperature, and Sampling  | ■ Reading  | 30 min · 75 XP  |
| 03 | Zero-Shot Prompting: Writing Effective Instructions  | ■ Reading  | 30 min · 75 XP  |
| 04 | Few-Shot Prompting: Examples and In-Context Learning | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Prompt Design Challenges        | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 06 | Chain-of-Thought Prompting: Step-by-Step Reasoning | ■ Reading  | 30 min · 75 XP  |
| 07 | Tree-of-Thought and Graph-of-Thought Prompting     | ■ Reading  | 35 min · 75 XP  |
| 08 | Self-Consistency: Sampling Multiple Paths          | ■ Reading  | 30 min · 75 XP  |
| 09 | Least-to-Most Prompting for Complex Tasks          | ■ Reading  | 30 min · 75 XP  |
| 10 | Reasoning Prompts Practice                         | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                |            |                 |
|----|------------------------------------------------|------------|-----------------|
| 11 | System Prompts: Persona, Tone, and Constraints | ■ Reading  | 30 min · 75 XP  |
| 12 | Role Prompting and Character Consistency       | ■ Reading  | 30 min · 75 XP  |
| 13 | Instruction Hierarchy and Prompt Priority      | ■ Reading  | 30 min · 75 XP  |
| 14 | Prompt Templates and Variable Injection        | ■ Reading  | 30 min · 75 XP  |
| 15 | System Prompt Design Practice                  | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                      |           |                 |
|----|------------------------------------------------------|-----------|-----------------|
| 16 | Structured Output: JSON, XML, and Schema Enforcement | ■ Reading | 35 min · 75 XP  |
| 17 | Markdown and Format Control in Outputs               | ■ Reading | 30 min · 75 XP  |
| 18 | Prompt Chaining: Connecting Multiple Steps           | ■ Reading | 35 min · 75 XP  |
| 19 | Retrieval-Augmented Prompting                        | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Prompt-Driven Content Pipeline   | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 21 | Evaluation: Measuring Prompt Quality and Consistency | ■ Reading  | 35 min · 75 XP  |
| 22 | A/B Testing Prompts: Systematic Improvement          | ■ Reading  | 35 min · 75 XP  |
| 23 | Prompt Injection: Attack Vectors and Defences        | ■ Reading  | 35 min · 75 XP  |
| 24 | Jailbreaking and Safety Considerations               | ■ Reading  | 30 min · 75 XP  |
| 25 | Evaluation and Safety Practice                       | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                            |            |                 |
|----|--------------------------------------------|------------|-----------------|
| 26 | Code Generation Prompts: Best Practices    | ■ Reading  | 35 min · 75 XP  |
| 27 | Data Extraction and Transformation Prompts | ■ Reading  | 35 min · 75 XP  |
| 28 | Summarisation Prompts for Long Documents   | ■ Reading  | 30 min · 75 XP  |
| 29 | Multi-Modal Prompting: Images and Audio    | ■ Reading  | 35 min · 75 XP  |
| 30 | Domain-Specific Prompt Practice            | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                          |           |                |
|----|----------------------------------------------------------|-----------|----------------|
| 31 | Cost Optimisation: Token Reduction Strategies            | ■ Reading | 30 min · 75 XP |
| 32 | Model-Specific Tuning: Claude, GPT-4, Gemini             | ■ Reading | 35 min · 75 XP |
| 33 | Prompt Management: Version Control and Libraries         | ■ Reading | 30 min · 75 XP |
| 34 | The Future of Prompting: DSPy and Automated Optimisation | ■ Reading | 35 min · 75 XP |



# MLOps & Model Deployment

■ Pending

Slug: **mlops** · Difficulty: **Advanced** · Duration: **36h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: MLflow, Kubeflow, Docker, Monitoring, Feature Stores

## Module 1

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 01 | Introduction to MLOps: Principles and the ML Lifecycle | ■ Reading  | 30 min · 75 XP  |
| 02 | Experiment Tracking with MLflow                        | ■ Reading  | 35 min · 75 XP  |
| 03 | Data Versioning with DVC                               | ■ Reading  | 35 min · 75 XP  |
| 04 | Feature Stores: Feast and Tecton                       | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: ML Experiment Pipeline            | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 06 | Model Training Pipelines with Kubeflow Pipelines    | ■ Reading  | 35 min · 75 XP  |
| 07 | Orchestrating ML Workflows with Airflow and Prefect | ■ Reading  | 35 min · 75 XP  |
| 08 | Hyperparameter Optimisation with Optuna             | ■ Reading  | 35 min · 75 XP  |
| 09 | Distributed Training: Horovod and PyTorch DDP       | ■ Reading  | 35 min · 75 XP  |
| 10 | Training Pipeline Practice: AutoML with Optuna      | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 11 | Model Packaging: Docker, ONNX, and Serialisation | ■ Reading  | 35 min · 75 XP  |
| 12 | Model Registry: MLflow, Vertex AI, and SageMaker | ■ Reading  | 35 min · 75 XP  |
| 13 | REST API Serving with FastAPI and TorchServe     | ■ Reading  | 35 min · 75 XP  |
| 14 | Batch Inference vs Online Inference              | ■ Reading  | 35 min · 75 XP  |
| 15 | Serving Practice: Model API with Autoscaling     | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                  |           |                 |
|----|--------------------------------------------------|-----------|-----------------|
| 16 | Model Monitoring: Data Drift and Concept Drift   | ■ Reading | 35 min · 75 XP  |
| 17 | Evidently AI: Monitoring Reports and Test Suites | ■ Reading | 35 min · 75 XP  |
| 18 | A/B Testing and Shadow Deployment                | ■ Reading | 35 min · 75 XP  |
| 19 | Feedback Loops and Continuous Training           | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: End-to-End MLOps Pipeline    | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 21 | CI/CD for ML: GitHub Actions and DVC Pipelines       | ■ Reading  | 35 min · 75 XP  |
| 22 | Infrastructure as Code for ML: Terraform + SageMaker | ■ Reading  | 35 min · 75 XP  |
| 23 | Kubernetes for ML: KServe and Seldon Core            | ■ Reading  | 35 min · 75 XP  |
| 24 | GPU Management and Resource Scheduling               | ■ Reading  | 35 min · 75 XP  |
| 25 | CI/CD and Infrastructure Practice                    | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 26 | Model Explainability in Production: SHAP and LIME       | ■ Reading  | 35 min · 75 XP  |
| 27 | Fairness Monitoring and Bias Detection                  | ■ Reading  | 35 min · 75 XP  |
| 28 | Cost Optimisation: Spot Instances and Model Compression | ■ Reading  | 35 min · 75 XP  |
| 29 | Security: Model Theft and Adversarial Attacks           | ■ Reading  | 35 min · 75 XP  |
| 30 | Governance and Security Practice                        | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                          |           |                |
|----|----------------------------------------------------------|-----------|----------------|
| 31 | LLMOps: Fine-Tuning, Evaluation, and Deployment at Scale | ■ Reading | 35 min · 75 XP |
| 32 | Vector Database Management and Embedding Pipelines       | ■ Reading | 35 min · 75 XP |
| 33 | Observability: Prometheus, Grafana, and ML Metrics       | ■ Reading | 35 min · 75 XP |
| 34 | Building an Internal ML Platform                         | ■ Reading | 35 min · 75 XP |



# Databases

## PostgreSQL Advanced

■ Pending

Slug: `postgresql` · Difficulty: **Intermediate** · Duration: **28h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Indexing, Query Optimization, Partitioning, JSONB

### Module 1

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 01 | Introduction to PostgreSQL: Architecture and Installation | ■ Reading  | 30 min · 75 XP  |
| 02 | Data Types: Numeric, Text, JSON, Arrays, and UUID         | ■ Reading  | 35 min · 75 XP  |
| 03 | Tables, Constraints, and Relationships                    | ■ Reading  | 35 min · 75 XP  |
| 04 | Basic SQL: SELECT, INSERT, UPDATE, DELETE                 | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Schema Design and Basic Queries      | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                      |            |                 |
|----|------------------------------------------------------|------------|-----------------|
| 06 | Joins: Inner, Left, Right, Full, and Cross           | ■ Reading  | 35 min · 75 XP  |
| 07 | Aggregations: GROUP BY, HAVING, and Window Functions | ■ Reading  | 35 min · 75 XP  |
| 08 | Subqueries, CTEs, and Recursive Queries              | ■ Reading  | 35 min · 75 XP  |
| 09 | Views and Materialised Views                         | ■ Reading  | 35 min · 75 XP  |
| 10 | Advanced Query Practice: Analytics Dashboard         | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                                          |            |                 |
|----|----------------------------------------------------------|------------|-----------------|
| 11 | Indexes: B-Tree, Hash, GIN, GiST, and BRIN               | ■ Reading  | 35 min · 75 XP  |
| 12 | Index Design: Partial, Composite, and Expression Indexes | ■ Reading  | 35 min · 75 XP  |
| 13 | EXPLAIN ANALYZE: Reading and Optimising Query Plans      | ■ Reading  | 35 min · 75 XP  |
| 14 | Table Partitioning: Range, List, and Hash                | ■ Reading  | 35 min · 75 XP  |
| 15 | Indexing and Query Optimisation Practice                 | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                   |           |                 |
|----|---------------------------------------------------|-----------|-----------------|
| 16 | Transactions: ACID, Isolation Levels, and Locking | ■ Reading | 35 min · 75 XP  |
| 17 | MVCC: How PostgreSQL Manages Concurrency          | ■ Reading | 35 min · 75 XP  |
| 18 | Deadlocks: Detection, Prevention, and Resolution  | ■ Reading | 35 min · 75 XP  |
| 19 | Full-Text Search with tsvector and tsquery        | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: High-Performance Analytics DB | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                            |            |                 |
|----|--------------------------------------------|------------|-----------------|
| 21 | JSONB: Querying, Indexing, and Performance | ■ Reading  | 35 min · 75 XP  |
| 22 | Stored Procedures and PL/pgSQL Functions   | ■ Reading  | 35 min · 75 XP  |
| 23 | Triggers: Auditing and Data Integrity      | ■ Reading  | 35 min · 75 XP  |
| 24 | pg_cron and Scheduled Maintenance Jobs     | ■ Reading  | 30 min · 75 XP  |
| 25 | Advanced Features Practice: Audit System   | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | Replication: Streaming, Logical, and Slot-Based     | ■ Reading  | 35 min · 75 XP  |
| 27 | High Availability with Patroni and pg_auto_failover | ■ Reading  | 35 min · 75 XP  |
| 28 | Connection Pooling with PgBouncer                   | ■ Reading  | 35 min · 75 XP  |
| 29 | Backup and Recovery: pg_dump, pg_basebackup, PITR   | ■ Reading  | 35 min · 75 XP  |
| 30 | Replication and HA Practice                         | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                                     |           |                |
|----|-----------------------------------------------------|-----------|----------------|
| 31 | PostgreSQL Security: Roles, Row-Level Security, SSL | ■ Reading | 35 min · 75 XP |
|----|-----------------------------------------------------|-----------|----------------|

|    |                                                |           |                 |
|----|------------------------------------------------|-----------|-----------------|
| 32 | Vacuum, Analyse, and Autovacuum Tuning         | ■ Reading | 35 min · 75 XP  |
| 33 | PostgreSQL with Docker and Kubernetes          | ■ Reading | 30 min · 75 XP  |
| 34 | Extensions: PostGIS, TimescaleDB, and pgvector | ■ Reading | 35 min · 75 XP  |
| 35 | Capstone: Production PostgreSQL Cluster        | ■ Project | 60 min · 150 XP |

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# MongoDB & Atlas

■ Pending

Slug: **mongodb** · Difficulty: **Intermediate** · Duration: **22h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Aggregation, Indexes, Atlas Search, Transactions

## Module 1

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 01 | Introduction to MongoDB: Document Model and Use Cases  | ■ Reading  | 30 min · 75 XP  |
| 02 | BSON Types, Documents, and Collections                 | ■ Reading  | 30 min · 75 XP  |
| 03 | CRUD Operations: insertOne, find, updateOne, deleteOne | ■ Reading  | 30 min · 75 XP  |
| 04 | Query Operators: Comparison, Logical, and Array        | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Product Catalogue with MongoDB    | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 06 | Schema Design: Embedding vs Referencing               | ■ Reading  | 35 min · 75 XP  |
| 07 | Data Modelling Patterns: Bucket, Subset, and Computed | ■ Reading  | 35 min · 75 XP  |
| 08 | Indexes: Single Field, Compound, and Multikey         | ■ Reading  | 35 min · 75 XP  |
| 09 | Text Indexes and Full-Text Search                     | ■ Reading  | 35 min · 75 XP  |
| 10 | Schema Design Practice: Social Network Model          | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                             |            |                 |
|----|-------------------------------------------------------------|------------|-----------------|
| 11 | Aggregation Pipeline: \$match, \$group, \$lookup, \$project | ■ Reading  | 35 min · 75 XP  |
| 12 | Aggregation: \$unwind, \$bucket, \$facet, and \$merge       | ■ Reading  | 35 min · 75 XP  |
| 13 | Atlas Search: Lucene-Powered Full-Text Search               | ■ Reading  | 35 min · 75 XP  |
| 14 | Atlas Vector Search: Semantic Search with Embeddings        | ■ Reading  | 35 min · 75 XP  |
| 15 | Aggregation Pipeline Practice: Analytics Report             | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                  |           |                 |
|----|--------------------------------------------------|-----------|-----------------|
| 16 | Transactions: Multi-Document ACID Transactions   | ■ Reading | 35 min · 75 XP  |
| 17 | Change Streams: Real-Time Data Monitoring        | ■ Reading | 35 min · 75 XP  |
| 18 | Validation with JSON Schema                      | ■ Reading | 35 min · 75 XP  |
| 19 | Time Series Collections                          | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Real-Time Analytics Platform | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 21 | Replication: Replica Sets and Elections         | ■ Reading  | 35 min · 75 XP  |
| 22 | Sharding: Architecture, Keys, and Zone Sharding | ■ Reading  | 35 min · 75 XP  |
| 23 | Atlas Global Clusters and Multi-Cloud           | ■ Reading  | 35 min · 75 XP  |
| 24 | Connection Pooling and Driver Configuration     | ■ Reading  | 35 min · 75 XP  |
| 25 | Replication and Sharding Practice               | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                            |            |                 |
|----|------------------------------------------------------------|------------|-----------------|
| 26 | Security: Authentication, RBAC, and Field-Level Encryption | ■ Reading  | 35 min · 75 XP  |
| 27 | Performance Tuning: Explain Plans and Index Strategies     | ■ Reading  | 35 min · 75 XP  |
| 28 | Backup and Restore: mongodump and Atlas Backups            | ■ Reading  | 30 min · 75 XP  |
| 29 | MongoDB with Node.js: Mongoose and Native Driver           | ■ Reading  | 35 min · 75 XP  |
| 30 | Security and Performance Practice                          | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                       |           |                |
|----|-------------------------------------------------------|-----------|----------------|
| 31 | MongoDB with Python: PyMongo and Motor                | ■ Reading | 35 min · 75 XP |
| 32 | Atlas Data Federation: Querying S3 and Atlas          | ■ Reading | 35 min · 75 XP |
| 33 | Atlas App Services: Triggers and Serverless Functions | ■ Reading | 35 min · 75 XP |
| 34 | Migrating from RDBMS to MongoDB                       | ■ Reading | 35 min · 75 XP |





# Software Engineering

## Data Structures & Algorithms

■ Pending

Slug: **dsa** · Difficulty: **Intermediate** · Duration: **60h planned** · **22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Arrays, Trees, Graphs, DP, Sorting, Interview Prep

### Module 1

|    |                                                        |            |                 |
|----|--------------------------------------------------------|------------|-----------------|
| 01 | Introduction to DSA: Complexity Analysis and Big-O     | ■ Reading  | 30 min · 75 XP  |
| 02 | Arrays: Operations, Sliding Window, and Two Pointers   | ■ Reading  | 35 min · 75 XP  |
| 03 | Strings: Manipulation, Hashing, and Pattern Matching   | ■ Reading  | 35 min · 75 XP  |
| 04 | Hash Maps and Hash Sets: Frequency and Lookup Problems | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Array and String Challenges       | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 06 | Stacks: Implementation, Monotonic Stack, and Applications | ■ Reading  | 35 min · 75 XP  |
| 07 | Queues, Deques, and Priority Queues                       | ■ Reading  | 35 min · 75 XP  |
| 08 | Linked Lists: Singly, Doubly, and Cycle Detection         | ■ Reading  | 35 min · 75 XP  |
| 09 | Recursion and the Recursion Tree Method                   | ■ Reading  | 35 min · 75 XP  |
| 10 | Linear Data Structures Practice                           | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                                           |            |                 |
|----|-----------------------------------------------------------|------------|-----------------|
| 11 | Binary Search: Standard, Rotated Arrays, and Search Space | ■ Reading  | 35 min · 75 XP  |
| 12 | Sorting Algorithms: Merge Sort, Quick Sort, Heap Sort     | ■ Reading  | 35 min · 75 XP  |
| 13 | Trees: Binary Trees, Traversals, and BST                  | ■ Reading  | 35 min · 75 XP  |
| 14 | Tree Problems: LCA, Diameter, and Path Sum                | ■ Reading  | 35 min · 75 XP  |
| 15 | Binary Search and Tree Practice                           | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                                    |           |                 |
|----|----------------------------------------------------|-----------|-----------------|
| 16 | Heaps and Priority Queues: Top-K and Median        | ■ Reading | 35 min · 75 XP  |
| 17 | Tries: Prefix Trees for String Problems            | ■ Reading | 35 min · 75 XP  |
| 18 | Graphs: Representation, BFS, and DFS               | ■ Reading | 35 min · 75 XP  |
| 19 | Graph Algorithms: Dijkstra, Bellman-Ford, and Prim | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Graph-Based Problem Set        | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 21 | Topological Sort and Cycle Detection in Graphs  | ■ Reading  | 35 min · 75 XP  |
| 22 | Union-Find: Disjoint Set Union                  | ■ Reading  | 35 min · 75 XP  |
| 23 | Dynamic Programming: Memoization and Tabulation | ■ Reading  | 35 min · 75 XP  |
| 24 | DP Patterns: Knapsack, LCS, and Coin Change     | ■ Reading  | 35 min · 75 XP  |
| 25 | Graph and DP Practice                           | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 26 | DP on Trees and Strings                           | ■ Reading  | 35 min · 75 XP  |
| 27 | Greedy Algorithms: Activity Selection and Huffman | ■ Reading  | 35 min · 75 XP  |
| 28 | Backtracking: Permutations, Subsets, and N-Queens | ■ Reading  | 35 min · 75 XP  |
| 29 | Bit Manipulation: XOR, Masks, and Bit Tricks      | ■ Reading  | 35 min · 75 XP  |
| 30 | Advanced Algorithms Practice                      | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                                     |           |                |
|----|-----------------------------------------------------|-----------|----------------|
| 31 | Interval Problems: Merge, Insert, and Meeting Rooms | ■ Reading | 35 min · 75 XP |
|----|-----------------------------------------------------|-----------|----------------|

|    |                                                        |           |                 |
|----|--------------------------------------------------------|-----------|-----------------|
| 32 | Math for DSA: GCD, Primes, and Modular Arithmetic      | ■ Reading | 35 min · 75 XP  |
| 33 | System Design for Interviews: Trade-offs and Estimates | ■ Reading | 35 min · 75 XP  |
| 34 | Mock Interview Strategies and Time Management          | ■ Reading | 30 min · 75 XP  |
| 35 | Capstone: 30-Problem Interview Preparation Set         | ■ Project | 60 min · 150 XP |

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# System Design

■ Pending

Slug: **system-design** · Difficulty: **Advanced** · Duration: **40h planned · 22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Scalability, CAP Theorem, Load Balancing, Caching

## Module 1

|    |                                                               |            |                 |
|----|---------------------------------------------------------------|------------|-----------------|
| 01 | Introduction to System Design: Scale, Trade-offs, and Process | ■ Reading  | 30 min · 75 XP  |
| 02 | Estimation: Back-of-the-Envelope Calculations                 | ■ Reading  | 30 min · 75 XP  |
| 03 | CAP Theorem and PACELC                                        | ■ Reading  | 35 min · 75 XP  |
| 04 | Load Balancing: Algorithms, L4 vs L7, and Health Checks       | ■ Reading  | 35 min · 75 XP  |
| 05 | Foundation Practice: Design a URL Shortener                   | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                                         |            |                 |
|----|---------------------------------------------------------|------------|-----------------|
| 06 | Caching: Strategies, Eviction Policies, and Redis       | ■ Reading  | 35 min · 75 XP  |
| 07 | Database Selection: SQL vs NoSQL Decision Framework     | ■ Reading  | 35 min · 75 XP  |
| 08 | Database Scaling: Sharding, Replication, and Federation | ■ Reading  | 35 min · 75 XP  |
| 09 | Consistent Hashing: Distributed Key Partitioning        | ■ Reading  | 35 min · 75 XP  |
| 10 | Caching and Database Practice: Design a Cache System    | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                                          |            |                 |
|----|----------------------------------------------------------|------------|-----------------|
| 11 | Message Queues: Kafka, RabbitMQ, and SQS Trade-offs      | ■ Reading  | 35 min · 75 XP  |
| 12 | Event-Driven Architecture and Event Sourcing             | ■ Reading  | 35 min · 75 XP  |
| 13 | API Design: REST, GraphQL, and gRPC at Scale             | ■ Reading  | 35 min · 75 XP  |
| 14 | CDN: Edge Caching and Content Delivery                   | ■ Reading  | 35 min · 75 XP  |
| 15 | Messaging and API Practice: Design a Notification System | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                                    |           |                 |
|----|----------------------------------------------------|-----------|-----------------|
| 16 | Design a Social Feed: Fan-Out on Write vs Read     | ■ Reading | 35 min · 75 XP  |
| 17 | Design a Search Engine: Inverted Index and Ranking | ■ Reading | 35 min · 75 XP  |
| 18 | Design a Video Streaming Platform                  | ■ Reading | 35 min · 75 XP  |
| 19 | Design a Ride-Sharing System                       | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Design Twitter/X at Scale      | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 21 | Rate Limiting: Token Bucket and Sliding Window        | ■ Reading  | 35 min · 75 XP  |
| 22 | Service Discovery and DNS in Microservices            | ■ Reading  | 35 min · 75 XP  |
| 23 | Distributed Locks and Leader Election                 | ■ Reading  | 35 min · 75 XP  |
| 24 | Distributed Transactions: 2PC and Saga                | ■ Reading  | 35 min · 75 XP  |
| 25 | Distributed Systems Practice: Design a Payment System | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                   |            |                 |
|----|---------------------------------------------------|------------|-----------------|
| 26 | Monitoring and Observability at Scale             | ■ Reading  | 35 min · 75 XP  |
| 27 | Disaster Recovery: RPO, RTO, and Multi-Region     | ■ Reading  | 35 min · 75 XP  |
| 28 | Security at Scale: Zero-Trust and DDoS Mitigation | ■ Reading  | 35 min · 75 XP  |
| 29 | Design a Key-Value Store from Scratch             | ■ Reading  | 35 min · 75 XP  |
| 30 | Resilience and Security Practice                  | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                  |           |                |
|----|--------------------------------------------------|-----------|----------------|
| 31 | Design a Distributed File System: HDFS and GFS   | ■ Reading | 35 min · 75 XP |
| 32 | Design a Recommendation System                   | ■ Reading | 35 min · 75 XP |
| 33 | Design a Real-Time Collaborative Editor          | ■ Reading | 35 min · 75 XP |
| 34 | Interview Framework: How to Structure Any Design | ■ Reading | 30 min · 75 XP |



# Mobile Development

## React Native

■ Pending

Slug: **react-native** · Difficulty: **Intermediate** · Duration: **36h planned · 22h lessons** · Lessons: **35** (■28 ■5 ■0 ■2) · Tags: Expo, Navigation, Animations, Native Modules

### Module 1

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 01 | Introduction to React Native and Expo              | ■ Reading  | 30 min · 75 XP  |
| 02 | Core Components: View, Text, Image, and ScrollView | ■ Reading  | 30 min · 75 XP  |
| 03 | Styling with StyleSheet: Flexbox on Mobile         | ■ Reading  | 35 min · 75 XP  |
| 04 | Handling Touch and Gestures                        | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Profile Card App              | ■ Exercise | 50 min · 100 XP |

### Module 2

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 06 | State Management with useState and useContext | ■ Reading  | 35 min · 75 XP  |
| 07 | Navigation with React Navigation v7           | ■ Reading  | 35 min · 75 XP  |
| 08 | Stack, Tab, and Drawer Navigators             | ■ Reading  | 35 min · 75 XP  |
| 09 | Passing Data Between Screens                  | ■ Reading  | 30 min · 75 XP  |
| 10 | Navigation Practice: Multi-Screen Task App    | ■ Exercise | 50 min · 100 XP |

### Module 3

|    |                                            |            |                 |
|----|--------------------------------------------|------------|-----------------|
| 11 | FlatList and SectionList: Performant Lists | ■ Reading  | 35 min · 75 XP  |
| 12 | TextInput, Forms, and Keyboard Handling    | ■ Reading  | 35 min · 75 XP  |
| 13 | Fetching Data with the Fetch API and Axios | ■ Reading  | 35 min · 75 XP  |
| 14 | Async Storage and Secure Store             | ■ Reading  | 35 min · 75 XP  |
| 15 | Data and Forms Practice: Contact Manager   | ■ Exercise | 50 min · 100 XP |

### Module 4

|    |                                               |           |                 |
|----|-----------------------------------------------|-----------|-----------------|
| 16 | Camera, Image Picker, and Media Library       | ■ Reading | 35 min · 75 XP  |
| 17 | Maps with react-native-maps                   | ■ Reading | 35 min · 75 XP  |
| 18 | Push Notifications with Expo Notifications    | ■ Reading | 35 min · 75 XP  |
| 19 | Background Tasks and Location Services        | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Location-Based Mobile App | ■ Project | 60 min · 150 XP |

### Module 5

|    |                                                  |            |                 |
|----|--------------------------------------------------|------------|-----------------|
| 21 | Animations with Reanimated 3 and Gesture Handler | ■ Reading  | 35 min · 75 XP  |
| 22 | State Management at Scale with Zustand           | ■ Reading  | 35 min · 75 XP  |
| 23 | React Query for Mobile: Caching and Sync         | ■ Reading  | 35 min · 75 XP  |
| 24 | Authentication: OAuth, Biometrics, and JWT       | ■ Reading  | 35 min · 75 XP  |
| 25 | Animations and State Practice                    | ■ Exercise | 50 min · 100 XP |

### Module 6

|    |                                                     |            |                 |
|----|-----------------------------------------------------|------------|-----------------|
| 26 | TypeScript with React Native                        | ■ Reading  | 35 min · 75 XP  |
| 27 | Testing: Jest and React Native Testing Library      | ■ Reading  | 35 min · 75 XP  |
| 28 | Performance: Hermes Engine and Bundle Optimisation  | ■ Reading  | 35 min · 75 XP  |
| 29 | Native Modules: Bridging JavaScript and Native Code | ■ Reading  | 35 min · 75 XP  |
| 30 | Testing and Performance Practice                    | ■ Exercise | 50 min · 100 XP |

### Module 7

|    |                                     |           |                |
|----|-------------------------------------|-----------|----------------|
| 31 | Expo EAS Build: Cloud Build Service | ■ Reading | 30 min · 75 XP |
|----|-------------------------------------|-----------|----------------|

|    |                                               |           |                 |
|----|-----------------------------------------------|-----------|-----------------|
| 32 | App Store and Google Play Submission Process  | ■ Reading | 30 min · 75 XP  |
| 33 | OTA Updates with Expo Updates                 | ■ Reading | 30 min · 75 XP  |
| 34 | Crash Reporting and Analytics with Sentry     | ■ Reading | 30 min · 75 XP  |
| 35 | Capstone: Production React Native Application | ■ Project | 60 min · 150 XP |

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# Flutter & Dart

■ Pending

Slug: flutter · Difficulty: Intermediate · Duration: 40h planned · 22h lessons · Lessons: 35 (■28 ■5 ■0 ■2) · Tags: Widgets, State Management, BLoC, Animations, Dart

## Module 1

|    |                                                       |            |                 |
|----|-------------------------------------------------------|------------|-----------------|
| 01 | Introduction to Flutter and Dart Programming Language | ■ Reading  | 30 min · 75 XP  |
| 02 | Dart Fundamentals: Types, Functions, and OOP          | ■ Reading  | 35 min · 75 XP  |
| 03 | Flutter Architecture: Widgets, Trees, and Rendering   | ■ Reading  | 35 min · 75 XP  |
| 04 | Core Widgets: Container, Row, Column, and Stack       | ■ Reading  | 30 min · 75 XP  |
| 05 | Foundation Practice: Flutter UI Component Library     | ■ Exercise | 50 min · 100 XP |

## Module 2

|    |                                            |            |                 |
|----|--------------------------------------------|------------|-----------------|
| 06 | Material Design 3 and Cupertino Widgets    | ■ Reading  | 35 min · 75 XP  |
| 07 | Stateless vs Stateful Widgets and setState | ■ Reading  | 35 min · 75 XP  |
| 08 | BuildContext and Widget Lifecycle          | ■ Reading  | 35 min · 75 XP  |
| 09 | Custom Widgets and Composition Patterns    | ■ Reading  | 35 min · 75 XP  |
| 10 | Widget Practice: Dashboard UI              | ■ Exercise | 50 min · 100 XP |

## Module 3

|    |                                               |            |                 |
|----|-----------------------------------------------|------------|-----------------|
| 11 | Navigation 2.0 and GoRouter                   | ■ Reading  | 35 min · 75 XP  |
| 12 | Forms, Validation, and TextEditingController  | ■ Reading  | 35 min · 75 XP  |
| 13 | HTTP Requests with dio and http Package       | ■ Reading  | 35 min · 75 XP  |
| 14 | JSON Serialisation with json_serializable     | ■ Reading  | 35 min · 75 XP  |
| 15 | Navigation and Data Practice: News Reader App | ■ Exercise | 50 min · 100 XP |

## Module 4

|    |                                               |           |                 |
|----|-----------------------------------------------|-----------|-----------------|
| 16 | State Management with Provider                | ■ Reading | 35 min · 75 XP  |
| 17 | Riverpod: Modern State Management for Flutter | ■ Reading | 35 min · 75 XP  |
| 18 | BLoC Pattern: Business Logic Component        | ■ Reading | 35 min · 75 XP  |
| 19 | GetX: All-in-One Solution                     | ■ Reading | 35 min · 75 XP  |
| 20 | Mid-Course Project: Weather App with Riverpod | ■ Project | 60 min · 150 XP |

## Module 5

|    |                                                 |            |                 |
|----|-------------------------------------------------|------------|-----------------|
| 21 | Animations: AnimatedWidget, AnimationController | ■ Reading  | 35 min · 75 XP  |
| 22 | Implicit Animations and Hero Transitions        | ■ Reading  | 35 min · 75 XP  |
| 23 | Custom Painters and Canvas API                  | ■ Reading  | 35 min · 75 XP  |
| 24 | Flutter Maps and Geolocation                    | ■ Reading  | 35 min · 75 XP  |
| 25 | Animations and Graphics Practice                | ■ Exercise | 50 min · 100 XP |

## Module 6

|    |                                                    |            |                 |
|----|----------------------------------------------------|------------|-----------------|
| 26 | Firebase Integration: Auth, Firestore, and Storage | ■ Reading  | 35 min · 75 XP  |
| 27 | Push Notifications with Firebase Cloud Messaging   | ■ Reading  | 35 min · 75 XP  |
| 28 | Local Storage: Hive and SharedPreferences          | ■ Reading  | 35 min · 75 XP  |
| 29 | Platform Channels: Accessing Native APIs           | ■ Reading  | 35 min · 75 XP  |
| 30 | Firebase and Platform Practice                     | ■ Exercise | 50 min · 100 XP |

## Module 7

|    |                                                       |           |                |
|----|-------------------------------------------------------|-----------|----------------|
| 31 | Testing: Widget Tests, Integration Tests, and Mocks   | ■ Reading | 35 min · 75 XP |
| 32 | Performance: const Widgets and Rendering Optimisation | ■ Reading | 35 min · 75 XP |
| 33 | Flavors: Multiple App Variants from One Codebase      | ■ Reading | 35 min · 75 XP |
| 34 | Deploying to App Store and Google Play with Fastlane  | ■ Reading | 30 min · 75 XP |





# Summary by Category

| Category                  | Courses | Complete | Total Lessons | Est. Hours |
|---------------------------|---------|----------|---------------|------------|
| ■ Programming Languages   | 6       | 0/6      | 210           | 204h       |
| ■ Frontend Development    | 5       | 0/5      | 175           | 152h       |
| ■ Backend Development     | 5       | 0/5      | 175           | 174h       |
| ■ Cloud Computing         | 1       | 0/1      | 35            | 50h        |
| ■ DevOps & Infrastructure | 3       | 0/3      | 105           | 94h        |
| ■ AI & Machine Learning   | 13      | 7/13     | 455           | 513h       |
| ■ Databases               | 2       | 0/2      | 70            | 50h        |
| ■ Software Engineering    | 2       | 0/2      | 70            | 100h       |
| ■ Mobile Development      | 2       | 0/2      | 70            | 76h        |
| TOTAL                     | 39      | 7/39     | 1365          | 1413h      |